

Design and Analysis of Various Filters



Cal Poly
Pomona

College of Engineering

Sarah Boyd, Nisarg Ahir, Juan Estrada

Dr. Moussavi

ETE 3101 Circuit Analysis II Laboratory

Lab 5

April 16, 2026

TABLE OF CONTENTS

1. ABSTRACT.....	[Page #3]
2. INTRODUCTION	[Page #3]
3. REVIEW.....	[Page #4]
3a.Sallen Key design process.....	[Page #4]
3b.Multiple feedback design process.....	[Page #8]
3c.Butterworth design process.....	[Page #10]
3d.Analysis of stable circuits.....	[Page #15]
4.METHODOLOGY.....	[Page #17]
5. CIRCUITS.....	[Page#17]
5a.Sallen Key BPF.....	[Page #17]
5a1.Determining component values.....	[Page #17]
5a2.Simulation.....	[Page #22]
5a3.Analysis.....	[Page #24]
5b. Multiple Feedback BPF.....	[Page #27]
5b1.Determining component values.....	[Page #27]
5b2.Simulation.....	[Page #33]
5b3.Analysis.....	[Page #34]
5c. Butterworth LPF.....	[Page #37]
5c1.Determining component values.....	[Page #37]
5c2.Simulation.....	[Page #39]
5c3.Analysis.....	[Page #40]
5d.Twin T BSF.....	[Page #41]
5d1.Determining component values.....	[Page #41]
5d2.Simulation.....	[Page #45]
5d3.Analysis.....	[Page #47]
6. DISCUSSION AND CONCLUSION	[Page #49]
7. REFERENCES.....	[Page #51]

ABSTRACT:

This experiment investigated the design and analysis of four second-order analog filter topologies: a Sallen Key band-pass filter, a multiple-feedback band-pass filter, a Butterworth unity-gain low-pass filter and a Twin T band-stop filter. For each configuration the transfer function $H(s)$ was derived analytically using discussed techniques, analysis of the transfer function and system was done. Component values were selected to achieve the desired dampening ratio value and frequency responses. The analytical results were validated through frequency domain simulations conducted in Multisim. Simulated magnitude and phase responses were compared against the calculated theoretical predictions to assess the accuracy. The results demonstrated close agreement between calculated and simulated responses, confirming correct filter implementation and expected behavior for each configuration.

INTRODUCTION:

Filters are circuits designed to pass signals within a desired frequency range while attenuating unwanted frequency components. They are commonly used in signal processing, communication systems, instrumentation and data-acquisition systems. Depending on the application filters may be designed to pass low frequencies, pass a selected band of frequencies, or reject a narrow range of frequencies. Filters can also be classified as passive or active. Their order affects the steepness of the frequency response beyond the cutoff frequency.

This report investigates the design and behavior of several second-order filter topologies, including Sallen Key band-pass filters, a multiple-feedback band-pass filter, a Butterworth

unity-gain low-pass filter and a Twin T band stop filter. Each circuit was designed with component selection based on the desired dampening ratio, and then analyzed using its transfer function to determine important characteristics such as cutoff frequency, center frequency, bandwidth, gain, poles, zeros, damping ratio and quality factor. The circuits were then simulated in Multisim to verify the expected frequency response and compare theoretical results with simulated behavior.

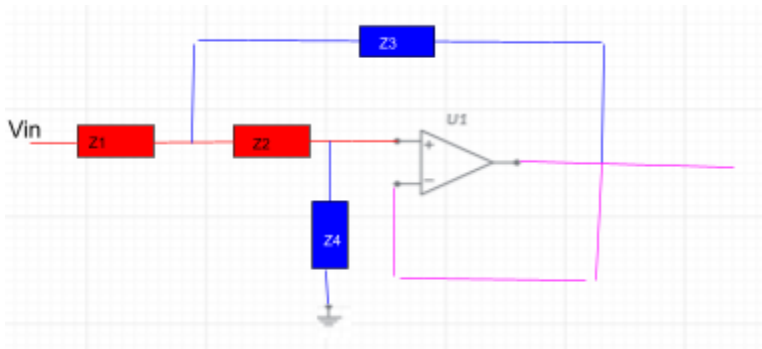
REVIEW:

This review section covers the design process for the three most useful filter topologies analyzed in this report: the Sallen-Key band-pass filter, the multiple-feedback band-pass filter and the Butterworth unity-gain low-pass filter.

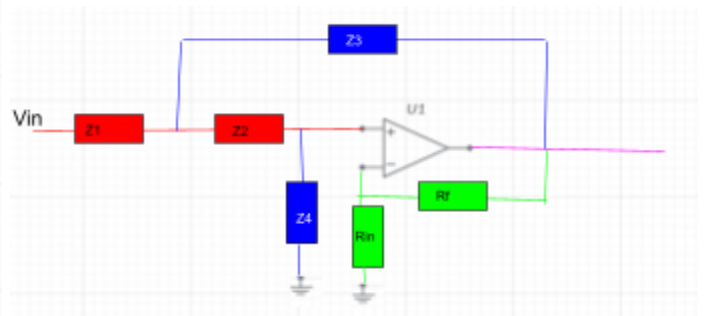
Sallen-Key Design Process:

A Sallen-Key filter is a second-order active filter topology that can be configured as a low-pass, high-pass or band-pass filter. In each case, the circuit uses an RC network with an operational amplifier. For second-order low-pass and high-pass Sallen-Key filters the two critical frequencies may be equal or different depending on the selected component values. If different cutoff frequencies are desired ($f_{c1} \neq f_{c2}$) One common design approach is to choose unequal resistor values while keeping the capacitors equal, keeping in mind that to increase the cutoff frequency we need to decrease the resistance values. This simplifies component selection while still allowing the cutoff frequencies to be adjusted.

The shortcut shown below can be used to determine the transfer function and component relationships without performing the full nodal analysis each time.



IMG 1.



IMG 2. Schematic with optional R_f and R_{in}

	Z1	Z2	Z3	Z4
Low pass	R1	R2	C1	C2
High pass	C1	C2	R1	R2

Table 1. Determining component values for the schematic in images 1 and 2.

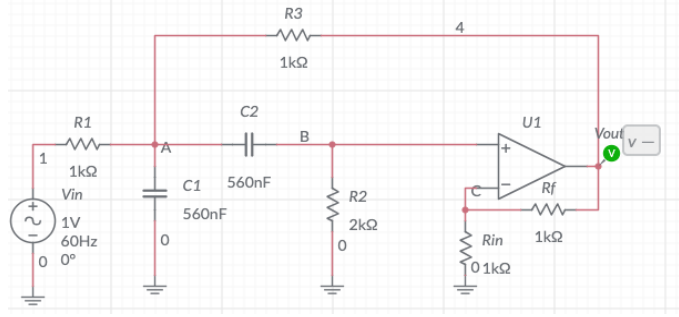
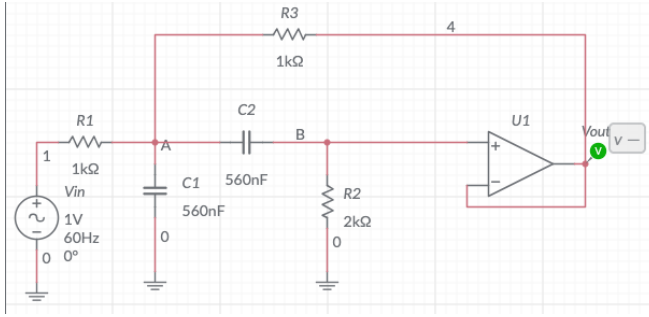
Pertaining to the transfer functions, there are two cases ($f_{c1} \neq f_{c2}$ or $f_{c1} = f_{c2}$) which influence the simplification due to resistor value implications. Once strategy is to recall the transfer functions for the case $f_{c1} \neq f_{c2}$ (noticing that the denominator remains the same) and then simplifying if $f_{c1} = f_{c2}$ is indeed the case.

	$f_{c1} = f_{c2}$	$f_{c1} \neq f_{c2}$
High pass	$H(s) = \frac{s^2}{s^2 + (\frac{2}{RC})s + (\frac{1}{RC})^2}$	$H(s) = \frac{s^2}{s^2 + (\frac{1}{R1C1} + \frac{1}{R2C2})s + \frac{1}{R1R2C1C2}}$
Low pass	$H(s) = \frac{(\frac{1}{RC})^2}{s^2 + (\frac{2}{RC})s + (\frac{1}{RC})^2}$	$H(s) = \frac{\frac{1}{R1R2C1C2}}{s^2 + (\frac{1}{R1C1} + \frac{1}{R2C2})s + \frac{1}{R1R2C1C2}}$

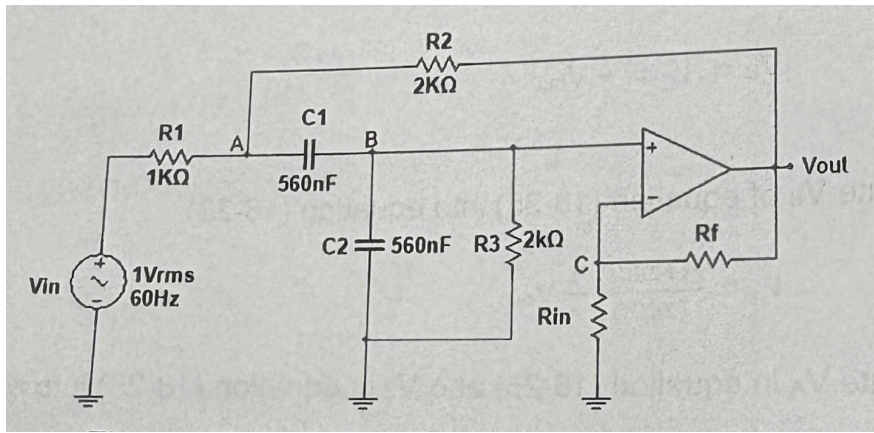
Table 2. Transfer functions for Sallen Key low and high pass filters.

For the equations shown in table 2, the op amp is assumed to act as a unity gain buffer meaning that $A=1$. This is shown in the schematic in image 1, as the output is connected directly back to the inverting input of the op amp. In image 2, the circuit includes optional feedback resistors R_f and R_{in} . The transfer function is the gain times the transfer function of the passive filter component, or $H(s)_{active} = AV \times H(s)_{passive}$. $H(s)$ is derived from the filter components of the circuit. The gain is $Av = -\frac{R_f}{R_{in}}$ for an inverting amplifier (input signal going into the negative terminal), and $Av = 1 + \frac{R_f}{R_{in}}$ for a non inverting amplifier (input signal going into the positive terminal).

For a Sallen Key bandpass, the process and circuit configuration varies.



IMG 3. Sallen Key BPF type one **IMG 4.** Type 1 Sallen Key BPF with the optional R_f and R_{in} components



IMG 5. Sallen Key BPF type two. This image was from Dr. Moussavi's book "Circuit Analysis in the Frequency Domain & Analog Filter Design". More information about this source can be found in the reference section of the report.

Type 1	Type 2
$R1 = R3 = R_{in} = R$	$C1 = C2 = C$
$R_f = (A - 1)R$	$m = \frac{1}{2\pi f_0 c_1}$
$R2 = 2R$	$R1 = \frac{2}{m}$
$C1 = C2 = C$	$R2 = \frac{1}{3R1}$

Type 1 transfer function:	$H(s) = \frac{\left(\frac{1}{RC}\right)S}{s^2 + \left(\frac{3-A}{RC}\right)S + \left(\frac{1}{RC}\right)^2}$
Type 2 transfer function:	$H(s) = \frac{\left(\frac{1}{RC^2}\right)S}{s^2 + \left(\frac{3-A}{RC}\right)S + \left(\frac{1}{RC}\right)^2}$

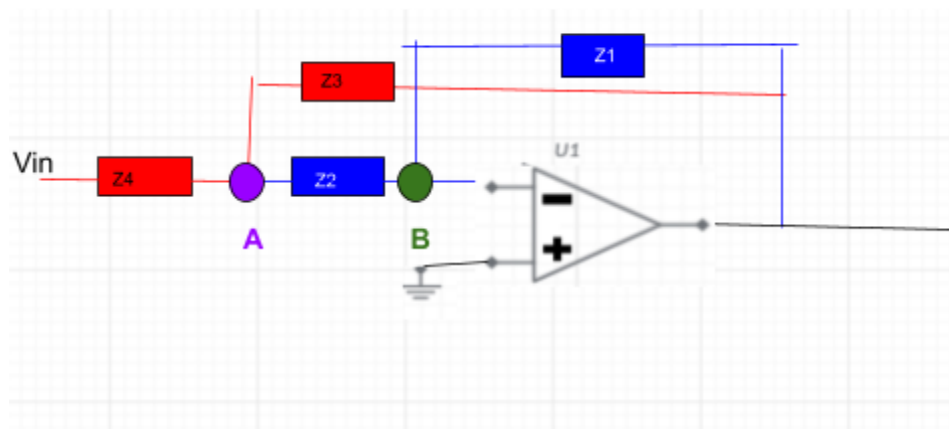
Table 4. Transfer functions for type 1 and 2. For type 1, note that $A = 1 + \frac{R_f}{R_{in}}$ and must be greater than 1 for a stable, underdamped circuit.

	$R3 = 2R1$
	$*Rf$

Table 3. Component equivalent values for type 2 are in order of selection. Additionally, Rf is selected and then using the gain value Rin can be determined ($A = 1 + \frac{Rf}{Rin}$).

Multiple feedback design process

The following modified version of the multiple feedback circuit can be used to simplify the process:



IMG 6. Schematic for multiple feedback design short cut

$Z1$ and $Z2$ form stage 1, while $Z3$ and $Z4$ form stage 2. Each stage can be configured to be a low pass or high pass stage. To understand the locations of components of stages 1 and 2, it is helpful to discuss the band pass circuit, similar to the one covered in this report.

	Z1	Z2	Z3	Z4
Band pass	R1	C1	C2	R2

Table 5. Configuration of components for band pass circuit

As shown in table 5, when $Z1 = R1$ and $Z2 = C1$, a circuit with low pass filter behavior is formed

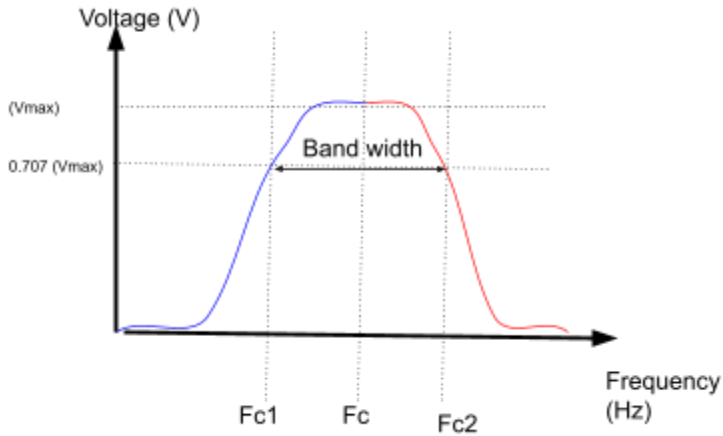
with $f_{c1} = \frac{1}{2\pi R1C1}$. When $Z3 = C2$ and $Z4 = R2$, a circuit with high pass filter behavior is formed

with $f_{c2} = \frac{1}{2\pi R2C2}$. For the band pass multiple feedback circuit we seemingly have the low pass

circuit before the high pass circuit despite a bandpass requiring stage 1 to be a high pass and

stage 2 to be low pass. This is the case because the high pass stage goes to the amplifier first,

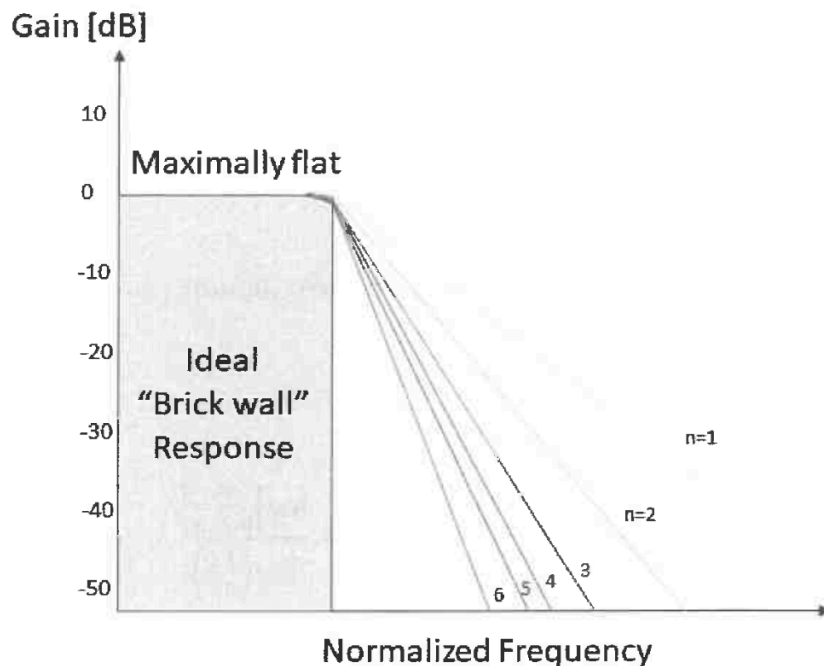
and then the low pass goes to the amplifier.



IMG 7. Frequency response for the multiple feedback circuit shown in image 6.

Butterworth Design Process:

Butterworth are used for when we want a flat passband with no ripple. Ideally a filter would be “like a brick wall” (seen in image 8) and drop straight down after the critical frequency, which is not the case for filters in real life and particularly for Butterworth and its slow transition rate from passband to stop band. Increasing the order of the filter can create a more severe attenuation past or before the critical frequency, as seen in image 8.

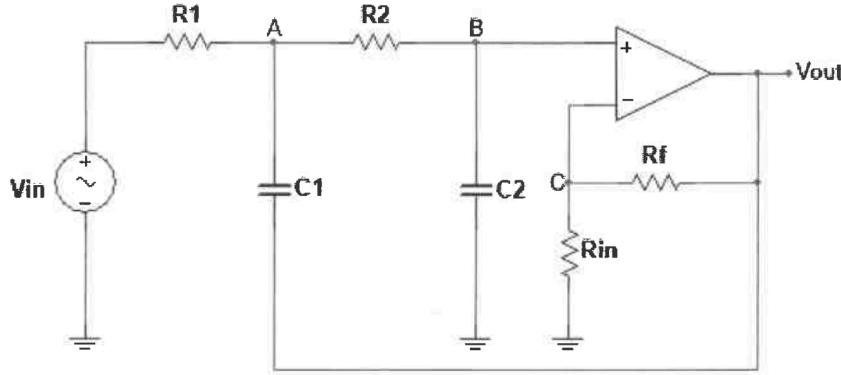


IMG 8. *The frequency response of a low pass Butterworth filter. This image is from chapter 19 of Dr. Moussavi's book " Circuit Analysis in the Frequency Domain & Analog Filter Design". More information about this source can be found in the reference section of the report.*

Butterworth gives a less steep attenuation after the cutoff frequency, which is acceptable as the flat passband is the most important aspect in this filter. Butterworth filter design following Sallen Key design topology is considered to be very popular due to being a maximally flat passband filter. Butterworth filters are used for audio, communication, control, ADC circuits and more. Pertaining to their use of circuits for conversion of analog to digital circuits, they can function as filters to get rid of aliases, and therefore filters for both digital and analog. According to the Nyquist Theorem, for accurate signal reconstruction of the frequency $f_{\text{sampling}} > 2f_{\text{maximum}}$ and for accurate signal reconstruction of the shape $f_{\text{sampling}} > 10f_{\text{maximum}}$, done to avoid an alias. However, this is the bare minimum and in practical scenarios a sampling frequency that is much greater than twice the maximum frequency present in the signal should be used.

The needed order of the filter can be calculated as $n \geq \frac{2.58}{\log(\frac{f_s}{f_p})} \cdot f_s$ (the stopband frequency, where the filter needs to strongly attenuate unwanted frequency) should be much larger than f_p (the passband frequency, or the highest frequency that should be passed through the filter with little attenuation). At minimum, f_s should be at least 10 times larger than f_p , so that $n \geq \frac{2.58}{\log(\frac{f_s}{f_p})}$ or $n \geq \frac{2.58}{\log(\frac{10f_p}{f_p})}$ or $n \geq \frac{2.58}{\log(10)}$ or 2.58 orders. Although 2.58 orders is the exact minimum, three orders is sufficient to meet this condition. Considering this equation is important when selecting the number of orders needed and considering what f_s is needed in order to meet this condition. It is important to keep in mind that the more orders added, the more poles are present, and the more complex the filter is.

For Butterworth, there are three cases critical to understand: limited gain, preselected gain, and unity gain.



IMG 9. Circuit for limited gain, second order low pass Butterworth filter. This image was from Dr. Moussavi's book "Circuit Analysis in the Frequency Domain & Analog Filter Design". More information about this source can be found in the reference section of the report.

Assuming that $f_{c1} = f_{c2}$ (the easiest and most preferable case), $R1=R2=R$ and $C1=C2=C$.

Using nodal analysis to find the transfer function would result in

$$H(s) = \frac{v_{out}}{v_{in}} = A \frac{\frac{1}{RC}}{s^2 + \frac{1}{RC}(3-A)s + \left(\frac{1}{RC}\right)^2} . \text{ Comparison to the standard second order denominator (}$$

$$\frac{constant}{s^2 + (2\xi\omega_n)s + \omega_n^2}) \text{ gives } \omega_n = \frac{1}{RC} . \text{ Using the table 6, the denominator is found to be } \frac{constant}{s^2 + \sqrt{2}s + 1}$$

,and since $\omega_n = \frac{1}{RC}$, the gain would therefore be $\omega_n(3 - A) = \sqrt{2}$. Gain can be solved for,

and since gain is $A = 1 + \frac{Rf}{Rin}$, Rf and Rin can be found to make the appropriate ratio.

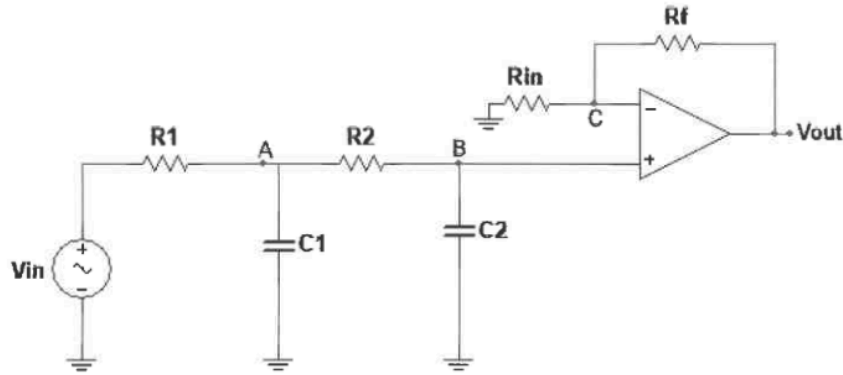
Order of the filter (n)	Normalized denominator polynomial (Ds)
1	$(s + 1)$
2	$(S^2 + \sqrt{2} S + 1)$
3	$(S^2 + 0.765s + 1)(S^2 + 1.848s + 1)$

Table 6. *The contents of this table are from Dr. Moussavi's book " Circuit Analysis in the Frequency Domain & Analog Filter Design". More information about this source can be found in the reference section of the report.*

Additionally, the resistance value needed for the desired dampening ratio can be solved for: $2\xi\omega_n = \omega_n(3 - A)$ so $\xi = \frac{3-A}{2}$. The process of analyzing the transfer function and finding poles and zeros continues as typical.

If gain is preselected, the process is similar and table 6 can be used, but there is a different denominator after doing nodal analysis. After doing nodal analysis, the transfer function would

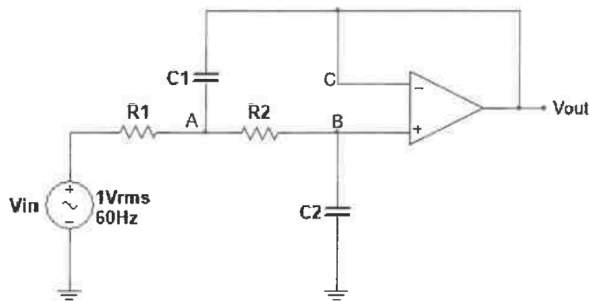
$$\text{be } H(s) = \frac{v_{out}}{v_{in}} = A \frac{\left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{3}{RC}\right)s + \left(\frac{1}{RC}\right)^2} \text{ or } A \frac{\frac{1}{RC}^2}{s^2 + \frac{1}{RC}(3+A)s + \left(\frac{1}{RC}\right)^2}$$



IMG 10. Circuit for preselected gain, second order low pass Butterworth filter. This image was from Dr. Moussavi's book "Circuit Analysis in the Frequency Domain & Analog Filter Design". More information about this source can be found in the reference section of the report.

For unity gain, nodal analysis gives a different denominator. Since $R_f=0$, $A=1$, since

$$A = 1 + \frac{R_f}{R_{in}} = 1 + \frac{0}{R_{in}} = 1$$

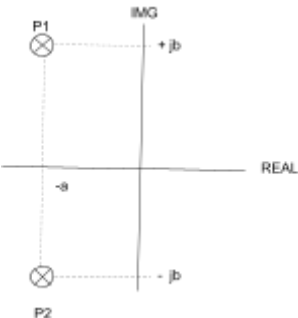
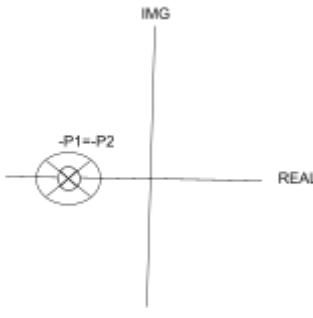
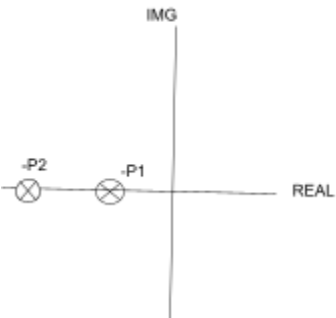
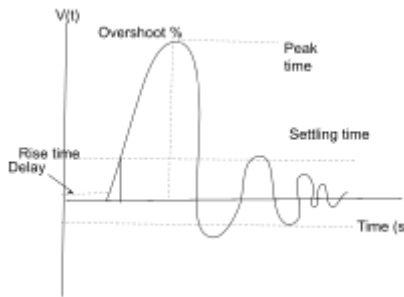
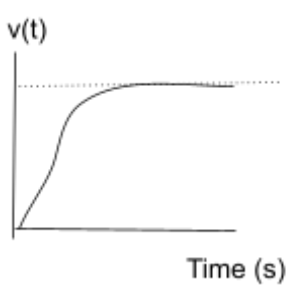
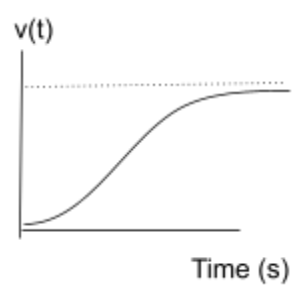


IMG 11. Circuit for unity gain, second order low pass Butterworth filter. This image was from Dr. Moussavi's book "Circuit Analysis in the Frequency Domain & Analog Filter Design". More information about this source can be found in the reference section of the report.

If $f_{c1} \neq f_{c2}$	If $f_{c1} = f_{c2}$
$R1 \neq R2$ $C1 = C2 = C$	$R1 = R2 = R$ $C1 = C2 = C$
$H(s) = \frac{\left(\frac{1}{R1R2C1C2}\right)}{s^2 + \left(\frac{1}{R1C} + \frac{1}{R2C}\right)s + \left(\frac{1}{R1R2C^2}\right)}$	$H(s) = \frac{\left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{2}{RC}\right)s + \left(\frac{1}{RC}\right)^2}$

Table 7. Resulting transfer functions for unity gain.

After obtaining the transfer function, analysis can be done:

Stable			
	Underdamped	Critically damped	Overdamped
	$0 < \xi < 1$	$\xi = 1$	$\xi > 1$
Poles	Two negative, complex poles $P_1 = -a + jb$ $P_2 = -a - jb$	Two identical, negative, real poles	Two negative, different, real poles
Stability			
Graph			
Behavior	Oscillating	Fast rise to steady state	Slower rise to steady state

Step Response	Underdamped: $V_o(t) = (1 - e^{\xi\omega_n t}(\cos(\omega_d t) + \frac{\xi}{\beta}\sin(\omega_d t))u(t)$
	Critically damped: $V_o(t) = (1 - [e^{-pt} + te^{-pt}])u(t)$
	Over damped : $V_o(t) = [1 - (Ae^{-p_1 t} + Be^{-p_2 t})]u(t)$

Table 8. For critically damped, when doing the quadratic equation to solve for poles the term under the square root must be equivalent to 0.

Regarding the equations for the graph for underdamped,

	Formula
Dampening factor	$\beta = \sqrt{1 - \xi^2}$
	$\omega_n = \quad -$
	$\omega_d = \beta\omega_n$
Rise time	$T_R = \frac{1 + 0.7\xi}{\omega_n}$
Delay time	$T_D = \frac{0.8 + 2.5\xi}{\omega_n}$
Peak time	$T_p = \frac{\pi}{\omega_d}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$

Table 9. β and ω_d should be calculated first, in order to find the other formulas. $\omega_o = \omega_n = \omega_r$, but NOT ω_d and ω_c .

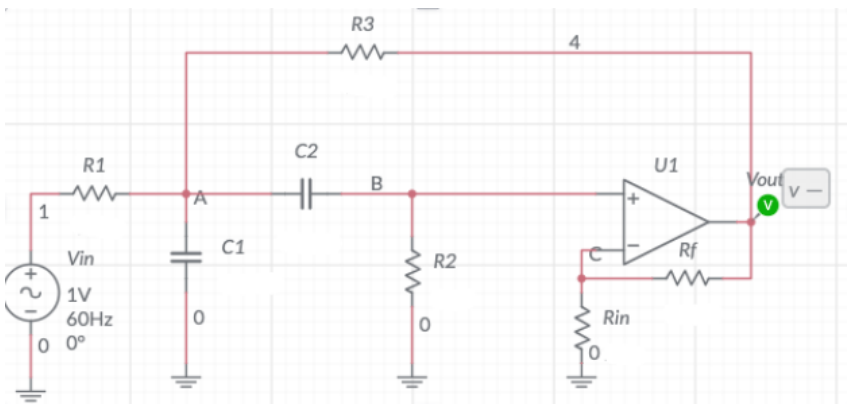
METHODOLOGY

The filter was first designed by selecting component values based on the desired damping ratio as well as the cutoff frequency. The circuit was then simulated to observe the expected frequency response. After simulation, calculations and analysis of the transfer function and theoretical circuit behavior was done. Finally, the calculated results were compared with the simulated results to evaluate how closely the theoretical response matched the observed simulated response.

CIRCUITS

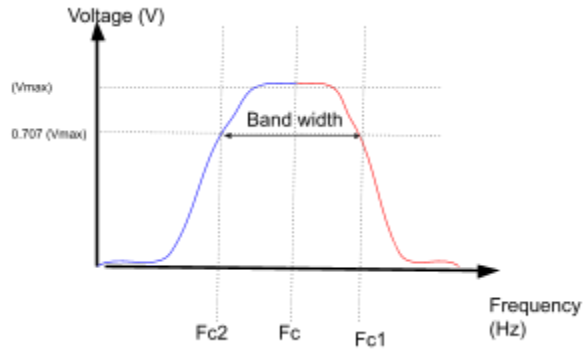
Circuit #1: Sallen Key Band Pass Filter Type 1 with R_f and R_{in} components

Choosing component values:



IMG 12. Sallen Key Band Pass Filter Type 1 with R_f and R_{in} components

The signal first passes through the filter consisting of C_2 and R_2 (stage 1) before reaching the op amp, hence its design as acting as a high pass filter. Because in a band pass filter the high pass filter section is a lower frequency due to it being the first stage (see image 13), we call $R_2=2R$.



Img 13. *The order of the cutoff frequencies is adjusted*

Since $F_{c1} \neq F_{c2}$, therefore $R_1 C_1 \neq R_2 C_2$ and $R_1 \neq R_2$ so that we can choose $C_1 = C_2 = 0.01\mu f$

To increase frequency we decrease R, so we choose $R_2 = 2R$ to make $f_{c1} > f_{c2}$ as in image 13.

Type 1
$R_1 = R_3 = R_{in} = R$
$R_f = (A - 1)R_{in}$
$R_2 = 2R$
$C_1 = C_2 = C = 0.01\mu f$

Table 10. *Component values*

Finding the transfer function:

Node C:

$$\frac{V_c - V_{ground}}{R_{in}} + \frac{V_c - V_{out}}{R_f} = 0$$

$$(R_f R_{in}) \frac{V_c - V_{ground}}{R_{in}} + (R_f R_{in}) \frac{V_c - V_{out}}{R_f} = 0$$

$$R_f(V_c) + R_{in}(V_c) - R_{in}(V_{out}) = 0$$

$$V_c(R_f + R_{in}) = R_{in}(V_{out})$$

V_c is the voltage at the inverting input feedback, so node C is just a voltage divider between out and ground. $V_c = V_{out} \frac{R_{in}}{R_f + R_{in}}$. Since $A = \frac{V_{out}}{V_c} = 1 + \frac{R_f}{R_{in}} = \frac{R_f + R_{in}}{R_{in}}$, $V_c = \frac{1}{A} V_{out}$

The voltage value at terminals of an op amp should be in balance, so therefore $V_b = V_c$ and

$$V_b = V_c = \frac{1}{A} V_{out}$$

$$\text{Node B: } \frac{V_b - V_a}{\frac{1}{C2S}} + \frac{V_b - V_{ground}}{R2} = 0$$

$$\frac{V_b - V_a}{\frac{1}{CS}} + \frac{V_b - 0}{2R} = 0$$

$$2RCS(V_b) - 2RCS(V_a) + V_b = 0$$

$$V_b(2RCS + 1) - 2RCS(V_a) = 0, \text{ so } V_b(2RCS + 1) = 2RCS(V_a)$$

Node B can give us V_a in terms of V_b . This is useful because V_b is written in terms of V_{out} , so we can write V_a in terms of v_{out} too, therefore giving us an equation with V_{in} and V_{out} .

$$V_a = V_b \left(\frac{2RCS + 1}{2RCS} \right) = \left(\frac{2RCS + 1}{2RCS} \right) \frac{1}{A} V_{out}$$

$$\text{Node A: } \frac{V_a - V_{in}}{R1} + \frac{V_a - V_{ground}}{\frac{1}{C1S}} + \frac{V_a - V_b}{\frac{1}{C2S}} + \frac{V_a - V_{out}}{R3} = 0$$

$$\frac{V_a - V_{in}}{R} + \frac{V_a - 0}{\frac{1}{CS}} + \frac{V_a - V_b}{\frac{1}{CS}} + \frac{V_a - V_{out}}{R} = 0$$

$$V_a - V_{in} + RCS(V_a) + RCS(V_a) - RCS(V_b) + V_a - V_{out} = 0$$

$$V_a(2RCS + 2) - V_{in} - RCS(V_b) - V_{out} = 0$$

$$2V_a(RCS + 1) = V_{in} + RCS(V_b) + V_{out}$$

$$\text{Since } V_a = \left(\frac{2RCS + 1}{2RCS} \right) \frac{1}{A} V_{out} \text{ and } V_b = V_c = \frac{1}{A} V_{out}$$

$$2 \left(\left(\frac{2RCS + 1}{2RCS} \right) \frac{1}{A} V_{out} \right) (RCS + 1) = V_{in} + RCS \left(\frac{1}{A} V_{out} \right) + V_{out}$$

$$\frac{2RCS + 1}{RCS} \frac{1}{A} V_{out} RCS + \frac{2RCS + 1}{RCS} \frac{1}{A} V_{out} = V_{in} + RCS \frac{1}{A} V_{out} + V_{out}$$

$$V_{out}(\frac{2RCS+1}{A} + \frac{2RCS+1}{ARCS}) = V_{in} + V_{out}(\frac{RCS}{A} + 1)$$

Plug in $x=RCS$

$$V_{out}(\frac{2x+1}{A} + \frac{2x+1}{Ax}) = V_{in} + V_{out}(\frac{x}{A} + 1)$$

$$V_{out}(\frac{2x^2+x}{Ax} + \frac{2x+1}{Ax}) = V_{in} + V_{out}(\frac{x}{A} + 1)$$

$$V_{out}(\frac{2x^2+x+2x+1}{Ax}) = V_{in} + V_{out}(\frac{x}{A} + 1)$$

$$V_{out}(\frac{2x^2+x+2x+1}{Ax}) = V_{in} + V_{out}(\frac{x}{A} + 1)$$

$$V_{out}(\frac{2x^2+x+2x+1}{Ax}) - V_{out}(\frac{x}{A} + 1) = V_{in}$$

$$V_{out}(\frac{2x^2+x+2x+1}{Ax} - \frac{x^2}{Ax} + x) = V_{in}, \text{ so } \frac{2x^2+x+2x+1}{Ax} - \frac{x^2}{Ax} + x = \frac{V_{in}}{V_{out}}$$

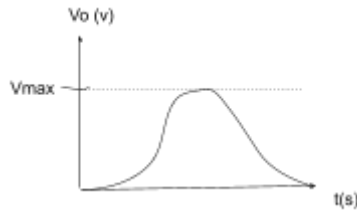
$$\frac{2x^2+x+2x+1-x^2+Ax}{Ax} = \frac{V_{in}}{V_{out}}, \frac{x^2+(3-A)+1}{Ax} = \frac{V_{in}}{V_{out}}, \text{ so } \frac{V_{out}}{V_{in}} = \frac{Ax}{x^2+(3-A)x+1} = \frac{A(RCS)}{RCS^2+(3-A)RCS+1}$$

After normalizing:

$$H(s) = \frac{V_{out}}{V_{in}} = A \frac{(\frac{1}{RC})s}{s^2 + (\frac{3-A}{RC})s + \frac{1}{RC}}, \text{ matching the type 1 transfer function in table 4.}$$

$$\text{As } S \rightarrow 0, H(s) = \frac{(\frac{1}{RC})s}{s^2 + (\frac{3-A}{RC})s + \frac{1}{RC}} \approx \frac{(\frac{1}{RC})0}{0^2 + (\frac{3-A}{RC})0 + \frac{1}{RC}} \approx \frac{0}{\frac{1}{RC}} \approx 0$$

$$\text{As } S \rightarrow \infty, H(s) = \frac{(\frac{1}{RC})s}{s^2 + (\frac{3-A}{RC})s + \frac{1}{RC}} \approx \frac{(\frac{1}{RC})\infty}{\infty^2 + (\frac{3-A}{RC})\infty + \frac{1}{RC}} \approx \frac{\infty}{\infty^2} \approx 0$$



Based on this analysis of the limits of s , the circuit attenuates low frequency and attenuates high frequency, thus **behaving like a band pass filter**.

Since the capacitance value is chosen, we need to now find the resistance value.

Compared $H(s) = A \frac{(\frac{1}{RC})s}{s^2 + (\frac{3-A}{RC})s + \frac{1}{RC}^2}$ to the standard second-order equation : $\frac{1}{s^2 + (2\xi)\omega_n + \omega_n^2}$

$$(2\xi)\omega_n = \frac{3-A}{RC}, \text{ so } (2\xi)\omega_n = (3 - A)\omega_n, \text{ and } \xi = \frac{(3-A)}{2}$$

Let's say that we want $\xi = 0.7$ for an underdamped system. $0.7 = \frac{(3-A)}{2}$, so $A = 1.6$

Recall that $A = 1 + \frac{Rf}{Rin}$, so $1.6 = 1 + \frac{Rf}{Rin}$, and $Rf = 0.6Rin$.

Let's say that we want $f_c = 22.5kHz$, $\omega_n = 2\pi f_c = 141.4k \text{ rad/sec}$

Recall that $C1 = C2 = 0.01f$, using a smaller capacitance value to make the resistance value large enough to be reasonable.

Using comparison of the second order equation, $\omega_n^2 = \frac{1}{RC}^2$, so $\omega_n = 141.4k \frac{\text{rad}}{\text{sec}} = \frac{1}{R(0.01 \text{ uf})}$

So $R = 707.2\Omega$. Since $Rf = 0.6Rin$, if $Rin=R=707.2\Omega$ $Rf = 424\Omega$.

Type 1
$R1 = R3 = Rin = R=707.2\Omega$
$Rf = (A - 1)Rin=424\Omega$
$R2 = 2R=1414\Omega$
$C1 = C2 = C=0.01\mu f$
$A=1.6$

Table 11.Adjusted component values

$$F_{c1} = f_0(\sqrt{1 + \xi^2} - \xi) = 22.5k(\sqrt{1 + (0.7)^2} - 0.7) = 11.7kHz$$

$$F_{c2} = f_0(\sqrt{1 + \xi^2} + \xi) = 22.5k(\sqrt{1 + (0.7)^2} + 0.7) = 43.2kHz$$

$$\text{Check: } f_c = \sqrt{f_2 f_1} = \sqrt{11.7k * 43.2k} = 22.48kHz$$

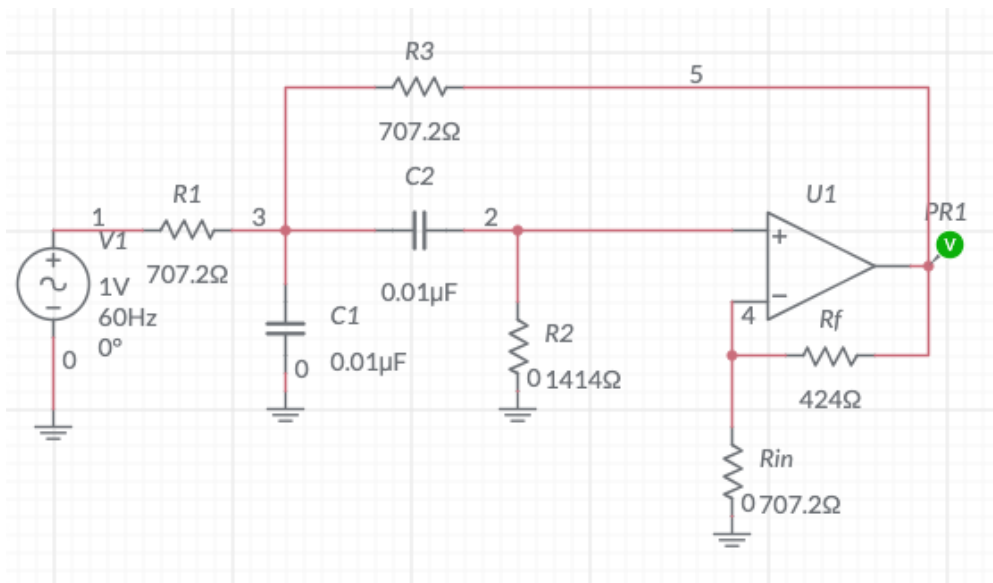
$$\text{Bandwidth} = 2\xi f_0 = 2(0.7)(22.5k) = 31.5kHz$$

$Q = \frac{1}{2\xi} = 0.714$, or $Q = \frac{f_c}{BW} = \frac{22.5\text{kHz}}{31.5\text{kHz}} = 0.714$, confirming we have an underdamped system

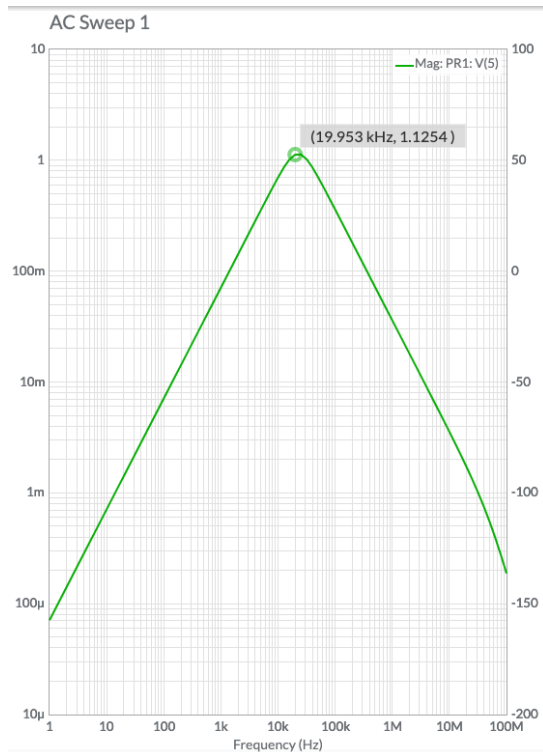
$Q > \frac{1}{2}$	Stable and underdamped
$Q = \frac{1}{2}$	Stable and critically damped
$Q < \frac{1}{2}$	Stable and overdamped

Table 12.

Simulation:

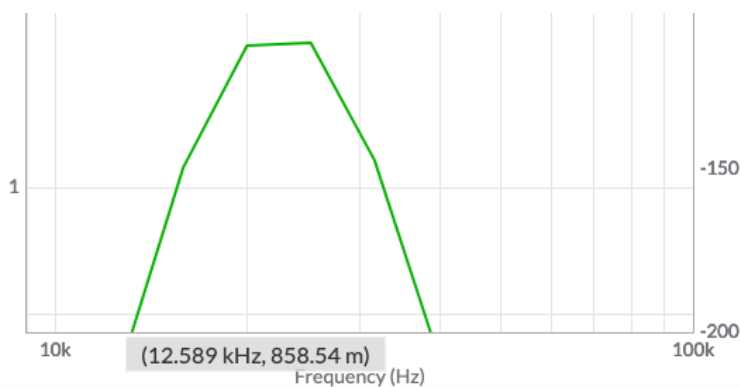


Img 14. Schematic used in the simulation. More information about the simulation software can be found in the reference section.

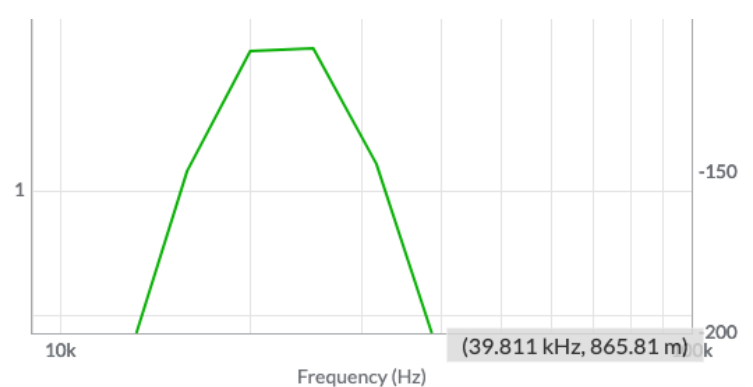


Img 15. Bode plot of the circuit. It can be seen that $f_c \approx 19.953\text{kHz}$

Finding the frequencies where the magnitude is 0.707 of the maximum voltage, or $0.707(1.254)$, gives us the upper and lower cutoff frequencies.



Img 16. F_{c2}



Img 17. F_{c1}

Results:

	fc1	fc2	fc	BW
Calculated	11.7k	43.2k	22.5k	31.5k
Measured	12.589k	39.8k	19.953k	27.9k

Table 13. Calculated versus measured (Multisim)

The % differences between the calculated and measured values were 7.60% for f_{c1} , 7.87% for f_{c2} , 11.32% for f_c and 11.43% for the bandwidth. This shows that the calculations and theoretical design are reasonably close to the simulated response.

Analysis:

Find poles and zeros:

$$H(s)_{total} = A \frac{(\frac{1}{RC})s}{s^2 + (\frac{3-A}{RC})s + \frac{1}{RC}}, \text{ so } N(Z) = 1.6 (S \frac{1}{RC}) \text{ and } N(P) = S^2 + (\frac{3-1.6}{2RC})S + \frac{1}{RC}$$

If $N(Z) = 0$, $0 = 1.6 (S \frac{1}{RC})$, so $S=0$ or $1.6 \frac{1}{RC} = 0$. Therefore there is one zero at zero ($S=0$) and one zero at infinity ($1.6 \frac{1}{RC} = 0$)

$$\text{If } N(P) = 0, 0 = S^2 + (\frac{3-1.6}{2RC})S + \frac{1}{RC}$$

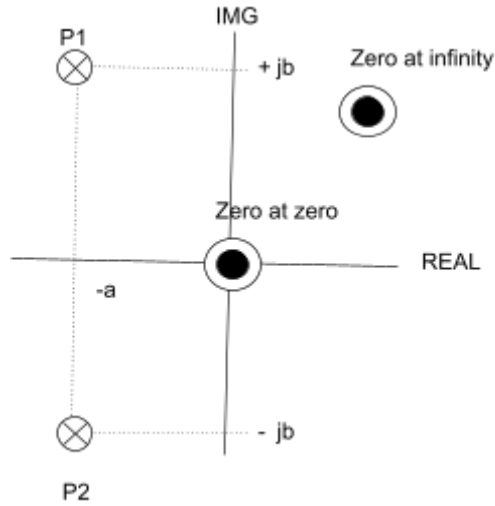
Using the quadratic formula ($\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1$, $b = \frac{1.4}{RC}$, $c = \frac{1}{R^2 C^2}$), we get

$$\frac{-\frac{1.4}{RC} \pm \sqrt{\frac{1.4^2}{RC^2} - 4(1)(\frac{1}{R^2 C^2})}}{2(1)} = \frac{-\frac{1.4}{RC} \pm \sqrt{\frac{1.96}{R^2 C^2} - 4(1)(\frac{1}{R^2 C^2})}}{2(1)} = \frac{-\frac{1.4}{RC} \pm \sqrt{\frac{-2.04}{R^2 C^2}}}{2(1)} = \frac{-\frac{1.4}{RC} \pm \frac{\sqrt{2.04}j}{RC}}{2} = -\frac{0.7}{RC} \pm j \frac{0.714}{RC}$$

$$\text{Pole 1: } S1 = -\frac{0.7}{RC} + j \frac{0.714}{RC}$$

$$\text{Pole 2: } S2 = -\frac{0.7}{RC} - j \frac{0.714}{RC}$$

The poles are negative and complex, meaning that the system is stable and underdamped.



IMG 18.

Since the system is underdamped, the following factors can be found.

	Formula	Calculation	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = \sqrt{1 - (0.7)^2}$	$\beta = 0.71$
	$\omega_d = \beta \omega_n$	$\omega_d = (0.71)(141.4k \frac{rad}{sec})$	$\omega_d = 100.98k \text{ rad/s}$
Rise time	$T_R = \frac{1 + 0.7\xi}{\omega_n}$	$T_R = \frac{1 + 0.7(0.7)}{14142.14}$	$T_R = 10.54\mu s$
Delay time	$T_D = \frac{0.8 + 2.5\xi}{\omega_n}$	$T_D = \frac{0.8 + 2.5(0.7)}{14142.14}$	$T_D = 18.03\mu s$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = \frac{\pi}{100.98k}$	$T_p = 31.11\mu s$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = e^{\frac{-\pi(0.71)}{0.71}} * 100$	$O.S\% = 4.60\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi \omega_n}$	For 2%: $t_s = \frac{-\ln(0.02)}{(0.7)(141.4k)}$	$t_s = 39.52\mu s$
		For 4%: $t_s = \frac{-\ln(0.04)}{(0.7)(141.4k)}$	$t_s = 32.51\mu s$
		For 5%: $t_s = \frac{-\ln(0.05)}{(0.7)(141.4k)}$	$t_s = 30.26\mu s$

Table 14. Underdamped system calculated values

The step response for an underdamped system is

$$V_o(t) = A [1 - e^{-\xi\omega_n t} (\cos(\omega_d t) + \frac{\xi}{\beta} \sin(\omega_d t))]$$

$$V_o(t) = 1.6[1 - e^{-(98.98k)t} (\cos 100.98kt) + \frac{0.7}{0.71} \sin(100.98kt)]$$

From the standard underdamped form, $e^{-\xi\omega_n t}$ so $-\xi\omega_n t = -\frac{t}{\tau}$ and $\tau = \frac{1}{-\xi\omega_n} = 10.10\mu s$

For 5τ , $5\tau = 5(10.10\mu s) = 50.5\mu s$

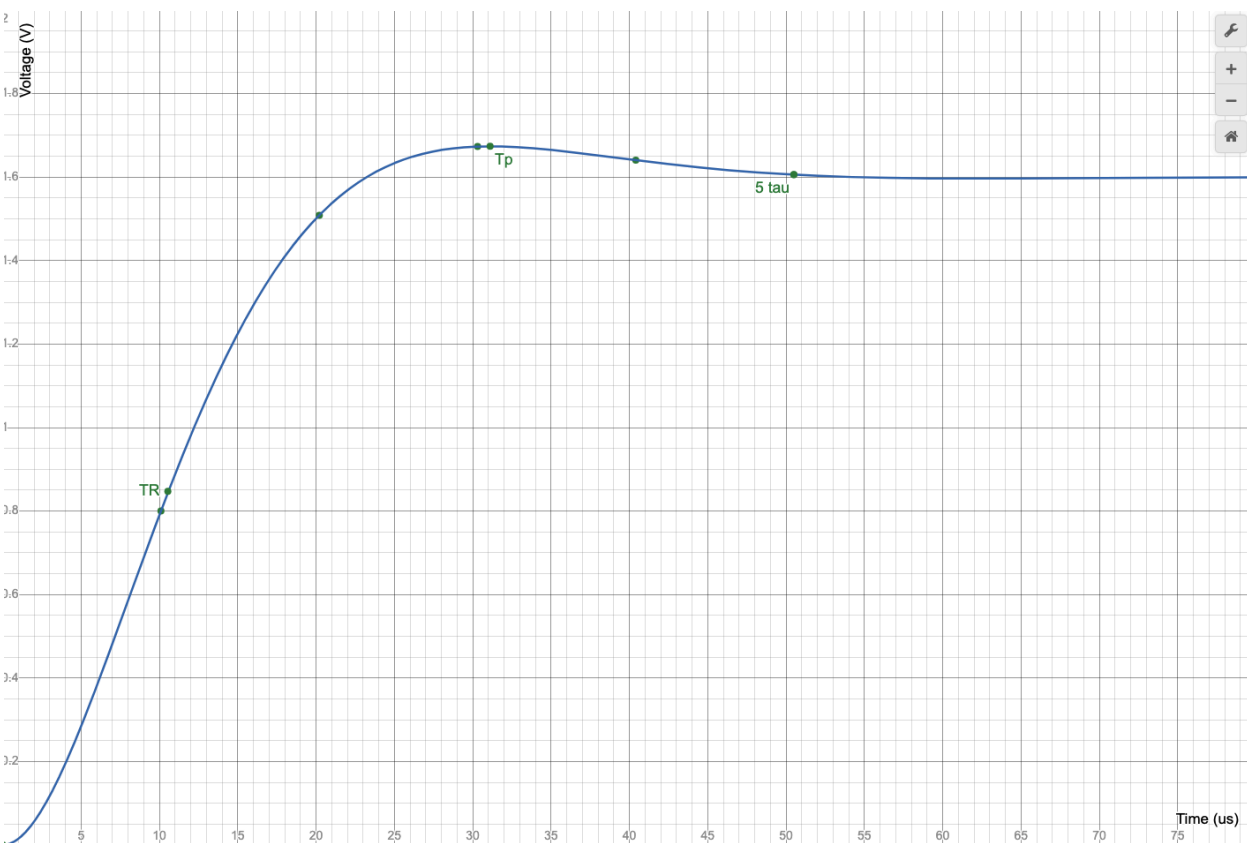
At $t = 5\tau$, $e^{-\xi\omega_n t} = e^{-\xi\omega_n (5\tau)} = e^{-5}$

$$V_o(t) = 1.6[1 - e^{-5} (\cos 100.98k(50.5\mu s) + 0.980 \sin(100.98k)(50.5\mu s))] = 1.6066v$$

To find the % of the final value that the output is, $\frac{1.606v}{1.6} * 100 = 100.36\%$

	$V_{out}(t)$	% Final
$0\tau=0s$	0v	0%
$1\tau = 10.10\mu s$	0.8004vv	50.02%
$2\tau = 20.21\mu s$	1.5087v	94.30%
$3\tau = 30.31\mu s$	1.6731v	104.57%
$4\tau = 40.41\mu s$	1.6405v	102.53%
$5\tau = 50.5\mu s$	1.6057v	100.36%

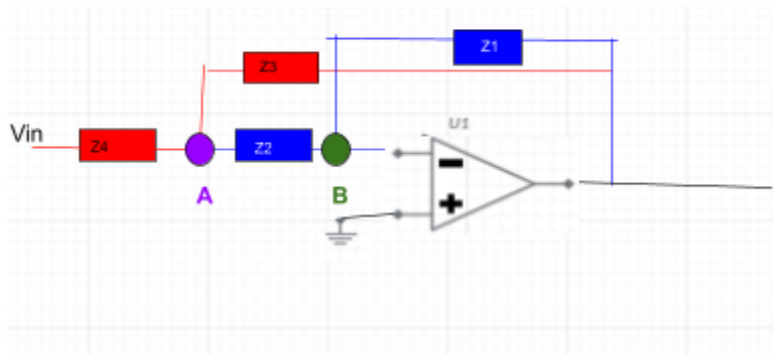
Table 15. $V_{out}(t)$ after time constants



IMG 19. Steady state is reached after 5τ , graphed using [Desmos.com](https://desmos.com). Note that the dampening ratio is larger meaning that the overshoot is smaller and making the overall graph shape much smoother.

Circuit #2: Multiple feedback BPF

1) Determining component values



IMG 20.

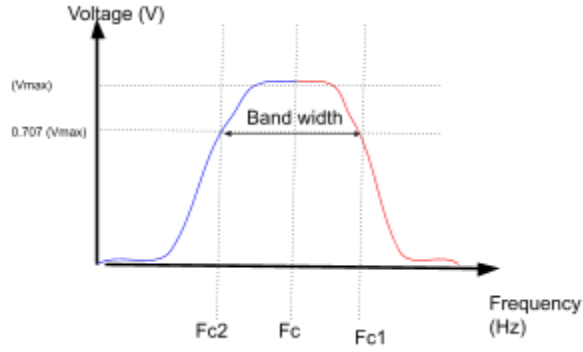
To make a bandpass circuit we need the following configuration:

	Z1	Z2	Z3	Z4
Band pass	R1	C1	C2	R2

Table 17.

Where $Z1 = R1$ and $Z2$ is $C1$, forming a low pass filter with the cutoff $F_{c1} = \frac{1}{2\pi R1C1}$

Where $Z3 = C2$ and $Z4 = R2$, forming a high pass filter with the cutoff $F_{c2} = \frac{1}{2\pi R2C2}$



Img 21. Note that the high pass filter with cutoff frequency f_{c2} is actually first. For an in depth explanation of why this is, please see the review section of the report.

Since $F_{c1} \neq F_{c2}$, therefore $R_1 C_1 \neq R_2 C_2$ and $R_1 \neq R_2$ so that we can choose $C_1 = C_2 = 0.1 \mu f$.

To increase frequency we decrease R, so we choose $R_2 = 2R_1$ to make $f_{c1} > f_{c2}$.

2) Finding the transfer function

Node A:

$$\frac{V_a - V_{in}}{R_2} + \frac{V_a - V_{out}}{\frac{1}{C_2 s}} + \frac{V_a - V_b}{\frac{1}{C_1 s}} = 0$$

Plug in equivalent values: $C_1 = C_2 = C$ and $R_2 = 2R$, and $R_1 = R$

$$\frac{V_a - V_{in}}{2R} + \frac{C s V_a - C s V_{out}}{1} + \frac{C s V_a - C s V_b}{1} = 0$$

$$V_a - V_{in} + 2RCS(V_a - V_{out}) + 2RCS(V_a - V_b) = 0$$

Node B:

$$\frac{V_b - V_a}{\frac{1}{C_1 s}} + \frac{V_b - V_{out}}{R_1} = 0$$

Plug in equivalent values: $C_1 = C_2 = C$ and $R2 = 2R$, and $R1 = R$

$$RCS(Vb - Va) + Vb - Vout = 0$$

Since $Vb = 0$

$$RCS(-Va) - Vout = 0$$

$$Va = \frac{-Vout}{RCS}$$

Using the equation from node A:

$$Va - Vin + 2RCS(Va - Vout) + 2RCS(Va - Vb) = 0$$

Plug in $Vb = 0$

$$Va - Vin + 2RCS(Va - Vout) + 2RCS(Va) = 0$$

Expand

$$Va - Vin + 2RCSVa - 2RCSVout + 2RCSVa = 0$$

Regroup

$$-Vin + Va(1 + 4RCS) - 2RCSVout = 0$$

$$Va(1 + 4RCS) - Vin - 2RCSVout = 0$$

Goal is to get $H(s) = \frac{Vout}{Vin}$, and by plugging in $Va = \frac{-Vout}{RCS}$ we can remove Va from the equation

$$\frac{-Vout}{RCS}(1 + 4RCS) - Vin - 2RCSVout = 0$$

$$\frac{-Vout}{RCS} + \frac{-Vout}{RCS}4RCS - Vin - 2RCSVout = 0$$

$$\frac{-Vout}{RCS} + -4Vout - Vin - 2RCSVout = 0$$

$$V_{out}[\frac{-1}{RCS} - 4 - 2RCS] - V_{in} = 0$$

$$\frac{V_{in}}{V_{out}} = [\frac{-1}{RCS} - 4 - 2RCS] \text{ so } \frac{v_{out}}{V_{in}} = \frac{1}{\frac{-1}{RCS} - 4 - 2RCS}$$

Simplifying:

$$\frac{1}{\frac{-1}{RCS} - 4 - 2RCS} \times \frac{RCS}{RCS} = \frac{-RCS}{1 + 4RCS + 2R^2C^2S^2} = \frac{-RCS}{(2R^2C^2)S^2 + (4RC)S + 1} = \frac{-\left(\frac{1}{2RC}\right)S}{S^2 + \left(-\frac{2}{2RC}\right)S + \frac{1}{(2R^2C^2)}}$$

Therefore

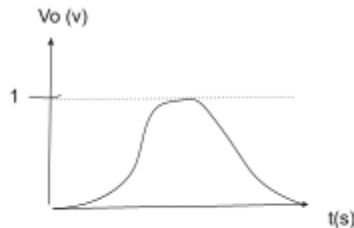
$$H(s) = A \frac{-\left(\frac{1}{2RC}\right)S}{S^2 + \left(-\frac{2}{2RC}\right)S + \frac{1}{(2R^2C^2)}}$$

To verify that this is indeed the transfer function of a band pass filter:

$$\text{As } S \rightarrow 0, H(s) = \frac{-\left(\frac{1}{2RC}\right)S}{S^2 + \left(-\frac{2}{2RC}\right)S + \frac{1}{(2R^2C^2)}} = \frac{0}{0 + 0 + \frac{1}{2R^2C^2}} = 0$$

$$\text{As } S \rightarrow \infty, H(s) = \frac{-\left(\frac{1}{2RC}\right)S}{S^2 + \left(-\frac{2}{2RC}\right)S + \frac{1}{(2R^2C^2)}} = \frac{\infty}{\infty^2 + \infty + \frac{1}{(2R^2C^2)}} \approx \frac{\infty}{\infty^2}$$

$$\text{Using L'Hopital's rule to account for } \frac{\infty}{\infty^2}, \frac{\frac{d}{ds} - \left(\frac{1}{2RC}\right)S}{\frac{d}{ds} \left(S^2 + \left(-\frac{2}{2RC}\right)S + \frac{1}{(2R^2C^2)}\right)} = \frac{-RC}{4R^2C^2S + 4RC} = 0$$



Based on this analysis of the limits of s, the circuit attenuates low frequency and attenuates high frequency, thus **behaving like a band pass filter**.

Compared to the standard second-order equation : $\frac{1}{s^2 + (2\xi)\omega_n + \omega_n^2}$

$$(2\xi)\omega_n = \frac{2}{2RC}$$

Let's say that we want $f_c = 22.5 \text{ kHz}$, $\omega_n = 2\pi f_c = 141.4 \text{ k rad/sec}$

Let's also say that we want $\xi = 0.7$ to have an underdamped system.

Recall that $C_1 = C_2 = 0.1 \mu\text{F}$

Therefore $2(0.7)(141.4 \text{ k}) = \frac{2}{2R(0.1 \mu\text{F})}$, the resistance value is $R = 50 \Omega$.

Our selected components are:

Impedances		Value
Z1	R1	=R=50
Z2	C1	=C=0.1 μF
Z3	C2	=C=0.1 μF
Z4	R2	=2R=100

Table 18. Selected components

The gain is $A = -\frac{R_2}{2R_1} = -\frac{100}{100} = -1$

Note that if we have a desired bandwidth, we can solve for the dampening ratio and then solve for R. If we have a desired dampening ratio, we can solve for R. For example,

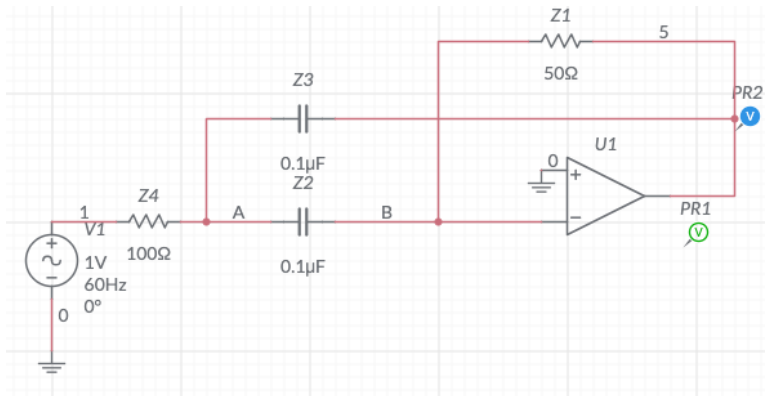
$Q = \frac{f_c}{BW} = \frac{f_c}{f_{c_2} - f_{c_1}} = \frac{\sqrt{f_2 f_1}}{f_2 - f_1}$ so we would plug in the bandwidth, solve for Q, and then find the

dampening ratio using $Q = \frac{1}{2\xi}$, so $\frac{1}{2\xi} = \frac{22.5 \text{ k}}{\text{our chosen BW}}$.

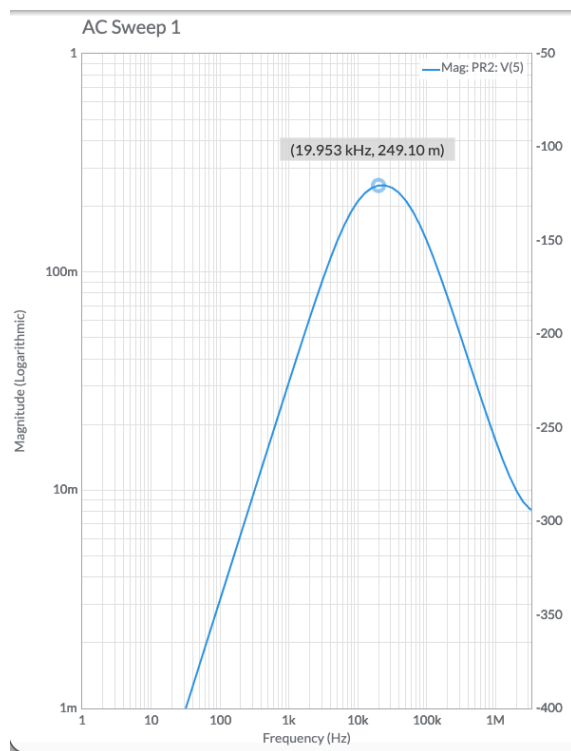
$$F_{c1} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi RC} = 31.83 \text{ kHz}$$

$$F_{c2} = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi 2RC} = 15.92 \text{ kHz}$$

Simulation:



Img 22. Schematic used in the simulation. More information about the simulation software can be found in the reference section.



Img 23. Based on the simulation, $f_c = 19.95\text{kHz}$. This is close to our theoretical f_c of 22.5kHz , More information about the simulation software can be found in the reference section.

Analysis:

Find poles and zeros:

$$H(s)_{total} = A \frac{-(\frac{1}{2RC})S}{S^2 + (\frac{2}{2RC})S + \frac{1}{2R^2C^2}}, \text{ so } N(Z) = - (S \frac{1}{2RC}) \text{ and } N(P) = S^2 + (\frac{2}{2RC})S + \frac{1}{2R^2C^2}$$

If $N(Z) = 0$, $0 = - (S \frac{1}{2RC})$, so $S=0$ or $-\frac{1}{2RC} = 0$. Therefore there is one zero at zero ($S=0$) and one zero at infinity ($-\frac{1}{2RC} = 0$)

$$\text{If } N(P) = 0, 0 = S^2 + (\frac{2}{2RC})S + \frac{1}{2R^2C^2}$$

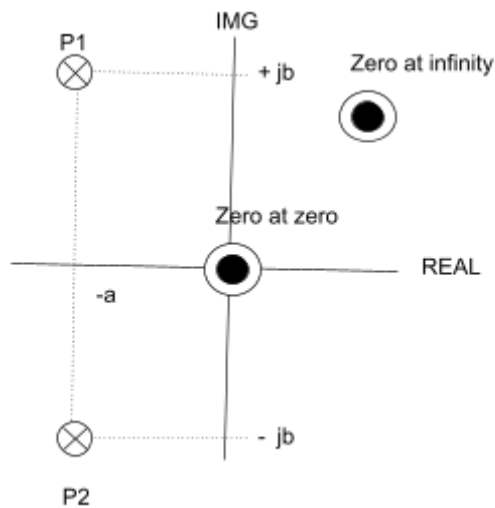
Using the quadratic formula ($\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1$, $b = \frac{1}{RC}$, $c = \frac{1}{2R^2C^2}$), we get

$$\frac{-\frac{1}{RC} \pm \sqrt{\frac{1}{RC}^2 - 4(1)(\frac{1}{2R^2C^2})}}{2(1)} = -\frac{1}{2RC} \pm j\frac{1}{2RC}$$

$$\text{Pole 1: } S_1 = -\frac{1}{2RC} + j\frac{1}{2RC}$$

$$\text{Pole 2: } S_2 = -\frac{1}{2RC} - j\frac{1}{2RC}$$

The poles are negative and complex, meaning that the system is stable and underdamped.



IMG 24.

Since the system is underdamped, the following factors can be found.

	Formula	Calculation	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = \sqrt{1 - (0.7)^2}$	$\beta = 0.71$
	$\omega_d = \beta \omega_n$	$\omega_d = (0.71)(14142.14)$	$\omega_d = 100.98k \text{ rad/s}$
Rise time	$T_R = \frac{1 + 0.7\xi}{\omega_n}$	$T_R = \frac{1 + 0.7(0.7)}{14142.14}$	$T_R = 10.54\mu\text{s}$
Delay time	$T_D = \frac{0.8 + 2.5\xi}{\omega_n}$	$T_D = \frac{0.8 + 2.5(0.7)}{14142.14}$	$T_D = 18.03\mu\text{s}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = \frac{\pi}{100.98k}$	$T_p = 31.11\mu\text{s}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = e^{\frac{-\pi(0.71)}{0.71}} * 100$	$O.S\% = 4.60\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi \omega_n}$	For 2%: $t_s = \frac{-\ln(0.02)}{(0.7)(141.4k)}$	$t_s = 39.52\mu\text{s}$
		For 4%: $t_s = \frac{-\ln(0.04)}{(0.7)(141.4k)}$	$t_s = 32.51\mu\text{s}$
		For 5%: $t_s = \frac{-\ln(0.05)}{(0.7)(141.4k)}$	$t_s = 30.26\mu\text{s}$

Table 19. Underdamped system calculated values

The step response for an underdamped system is $V_o(t) = A [1 - e^{-\xi \omega_n t} (\cos(\omega_d t) + \frac{\xi}{\beta} \sin(\omega_d t))]$

$$V_o(t) = -1 [1 - e^{-(98.98k)t} (\cos 100.98kt) + \frac{0.7}{0.71} \sin(100.98kt)]$$

From the standard underdamped form, $e^{-\xi \omega_n t}$ so $-\xi \omega_n t = -\frac{t}{\tau}$ and $\tau = \frac{1}{-\xi \omega_n} = 10.10\mu\text{s}$

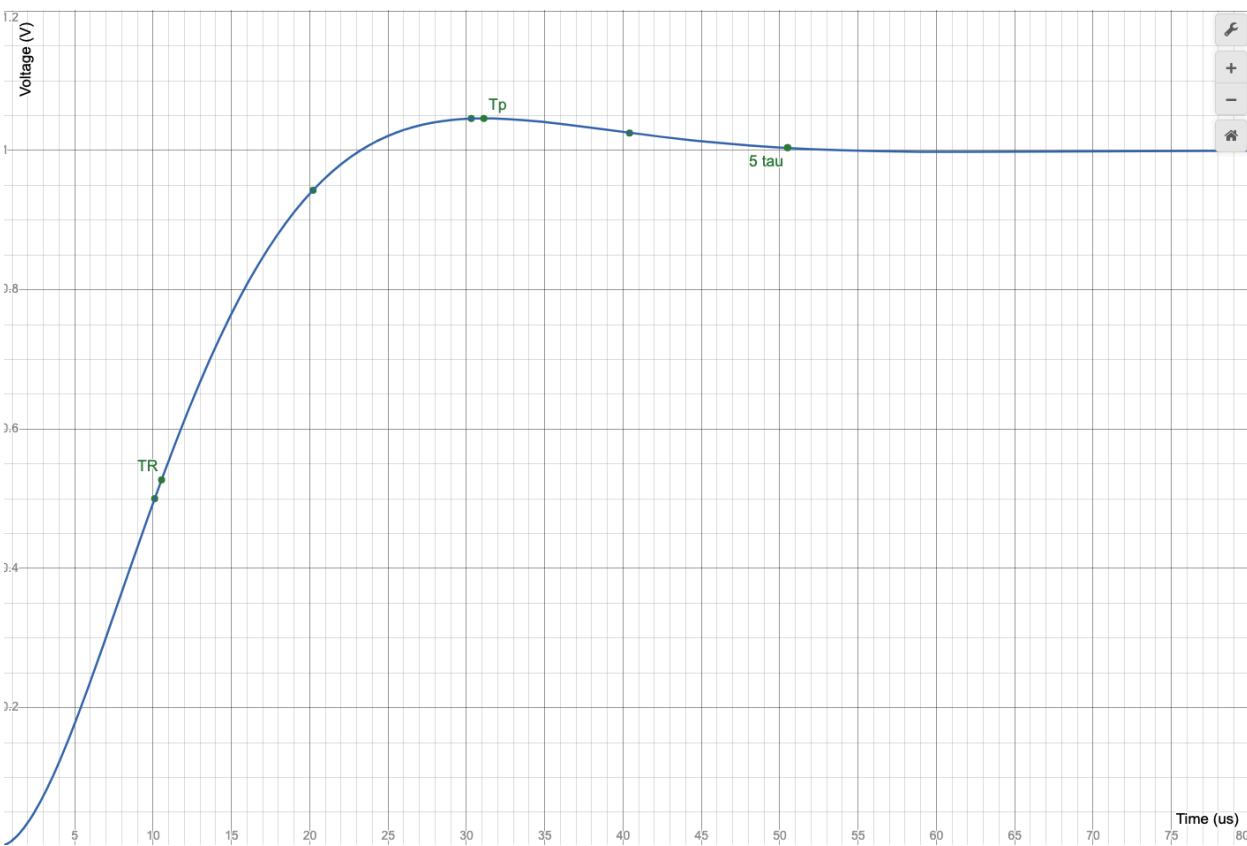
For 5τ , $5\tau = 5(10.10\mu\text{s}) = 50.5\mu\text{s}$. At $t = 5\tau$, $e^{-\xi \omega_n t} = e^{-\xi \omega_n (5\tau)} = e^{-5}$

$$V_o(t) = -1 [1 - e^{-5} (\cos 100.98k(50.5\mu\text{s}) + 0.980 \sin(100.98k)(50.5\mu\text{s}))] = -0.5002v$$

To find the % of the final value that the output is, $\frac{-0.5002v}{1} * 100 = 50.02\%$

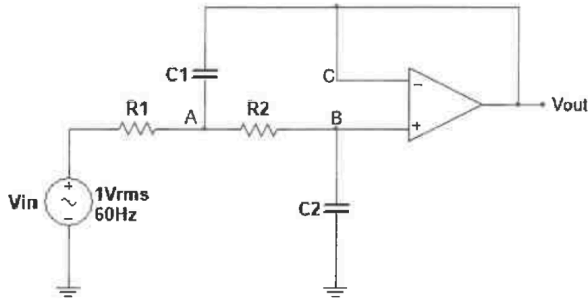
	$V_{out}(t)$	% Final
$0\tau=0s$	0v	0%
$1\tau = 10.10\mu s$	-0.5002v	50.02%
$2\tau = 20.21\mu s$	-0.9430v	94.30%
$3\tau = 30.31\mu s$	-1.0457v	104.57%
$4\tau = 40.41\mu s$	-1.0253v	102.53%
$5\tau = 50.5\mu s$	-1.0036v	100.36%

Table 15. $V_{out}(t)$ after time constants



IMG 25. Steady state is reached after 5τ , graphed using [Desmos.com](https://desmos.com). Note that the magnitude is graphed ($A=1$), but for this circuit $A=-1$. Note that the dampening ratio is larger meaning that the overshoot is smaller and making the overall graph shape much smoother.

Circuit #3: Butterworth unity gain low pass filter



IMG 26. Circuit for unity gain, second order low pass Butterworth filter. This image was from Dr. Moussavi's book "Circuit Analysis in the Frequency Domain & Analog Filter Design". More information about this source can be found in the reference section of the report.

Component selection:

First, the transfer function is found.

Node C: $V_c = V_{out}$

The voltage value at terminals of an op amp should be in balance, so therefore $V_b = V_c$ and $V_b = V_c = V_{out}$

Node B: $\frac{V_b - V_a}{R_2} + \frac{V_b - V_{ground}}{\frac{1}{C_2S}} = 0$, $V_b - V_a + R_2 C_2 S V_b = 0$, $V_a = (1 + R_2 C_2 S) V_b$

$V_a = (1 + R_2 C_2 S) V_b$

Node A: $\frac{V_a - V_{in}}{R_1} + \frac{V_a - V_c}{\frac{1}{C_1S}} + \frac{V_a - V_b}{R_2} = 0$,

$R_2 V_a - R_2 V_{in} + R_1 R_2 C_1 S (V_a) - R_1 R_2 C_1 S (V_{out}) + R_1 V_a - R_1 V_b = 0$

$V_a (R_2 + R_1 R_2 C_1 S + R_1) - R_1 V_b - R_2 V_{in} - R_1 R_2 C_1 S (V_{out}) = 0$

$V_a (2 + R_2 C_1 S) - V_{in} - (1 + R_2 C_1 S) V_{out} = 0$

$V_a = (1 + R_2 C_1 S) V_{out}$, so

$(1 + R_2 C_1 S) V_{out} (2 + R_2 C_1 S) - V_{in} - (1 + R_2 C_1 S) (V_{out}) = 0$

$V_{out} ((1 + R_2 C_1 S) (2 + R_2 C_1 S) - (1 + R_2 C_1 S)) = V_{in}$

$V_{out} ((1 + R_2 C_1 S) (2 + R_2 C_1 S - 1)) = V_{in}$

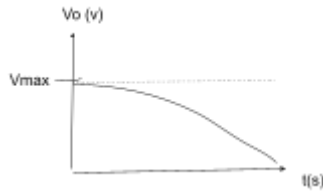
$V_{out} (1 + R_2 C_1 S)^2 = V_{in}$

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{1}{(1+RCs)^2} = \frac{1}{1+2RCs+(RCs)^2}$$

$$H(s) = \frac{\left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{2}{RC}\right)s + \left(\frac{1}{RC}\right)^2}$$

$$\text{As } s \rightarrow 0, H(s) = \frac{\left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{2}{RC}\right)s + \left(\frac{1}{RC}\right)^2} \approx \frac{\left(\frac{1}{RC}\right)^2}{0^2 + \left(\frac{2}{RC}\right)0 + \left(\frac{1}{RC}\right)^2} \approx \frac{\frac{1}{RC^2}}{\frac{1}{RC^2}} \approx 1$$

$$\text{As } s \rightarrow \infty, H(s) = \frac{\left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{2}{RC}\right)s + \left(\frac{1}{RC}\right)^2} \approx \frac{\left(\frac{1}{RC}\right)^2}{\infty^2 + \left(\frac{2}{RC}\right)\infty + \left(\frac{1}{RC}\right)^2} \approx \frac{\frac{1}{RC^2}}{\infty} \approx 0$$



Based on this analysis of the limits of s , the circuit attenuates high frequency, thus **behaving like a low pass filter**.

For unity gain, since $R_f=0$, $A = 1 + \frac{R_f}{R_{in}} = 1 + \frac{0}{R_{in}} = 1$. Through comparison to

$\frac{1}{s^2 + (2\xi\omega_n)s + \omega_n^2}$, $\omega_n = \frac{1}{RC}$. since $2\xi\omega_n = \frac{2}{RC}$, $2\xi = 2\omega_n$, so $\xi = 1$ and the circuit is critically

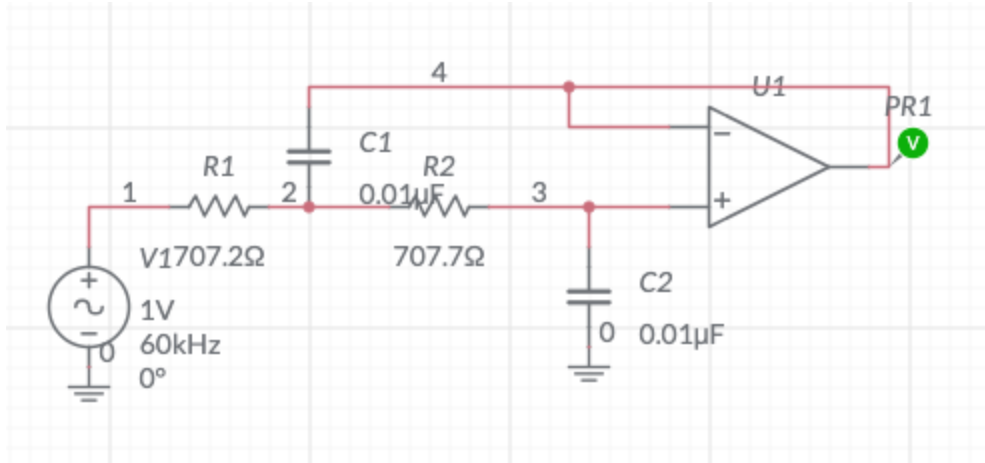
dampened. Let's say that we want $f_c = 22.5\text{khz}$, $\omega_n = 2\pi f_c = 141.4\text{k rad/sec}$. Recall

that $C1 = C2 = 0.01\mu\text{f}$, using a smaller capacitance value to make the resistance value large

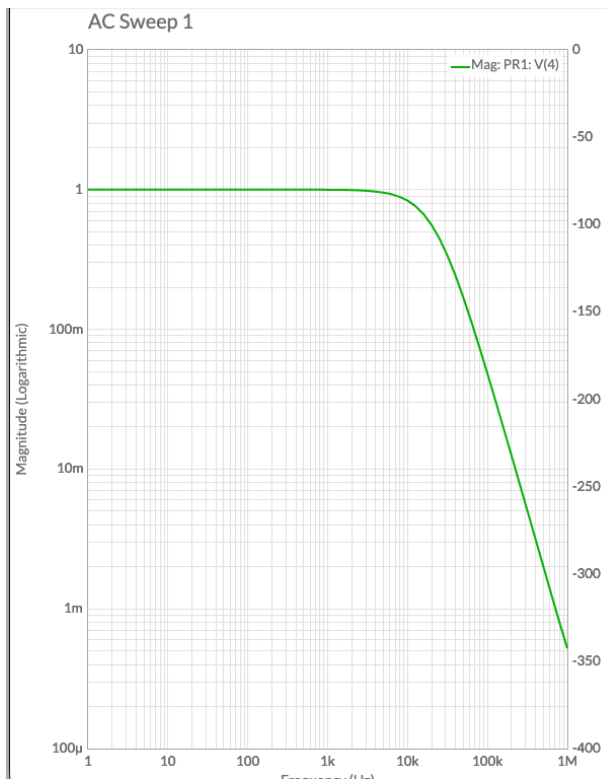
enough to be reasonable. Therefore, $2\xi\omega_n = \frac{2}{RC}$, $2(1)(141.4\text{k}) = \frac{2}{R(0.01\mu\text{f})}$, so $R=707.2\Omega$

R1	=R=707.2Ω
R2	=R=707.2Ω
C1	=C=0.01μf
C2	=C=0.01μf
A	=0

Table 16.Component values



Img 27. Schematic used in the simulation. More information about the simulation software can be found in the reference section.



Img 28. Bode plot from the simulation. More information about the simulation software can be found in the reference section.

Find poles and zeros

$$H(s)_{total} = H(s) = \frac{(\frac{1}{RC})^2}{s^2 + (\frac{2}{RC})s + (\frac{1}{RC})^2}, \text{ so } N(Z) = (\frac{1}{RC})^2 \text{ and } N(P) = s^2 + (\frac{2}{RC})s + \frac{1}{RC^2}$$

If $N(Z) = 0$, $0 = (\frac{1}{RC})^2$, so $\frac{1}{RC} = 0$ and $\frac{1}{RC} = 0$. Therefore we have two zeros at infinity, indicating a low pass filter of second order.

$$\text{If } N(P) = 0, 0 = s^2 + (\frac{2}{RC})s + (\frac{1}{RC})^2$$

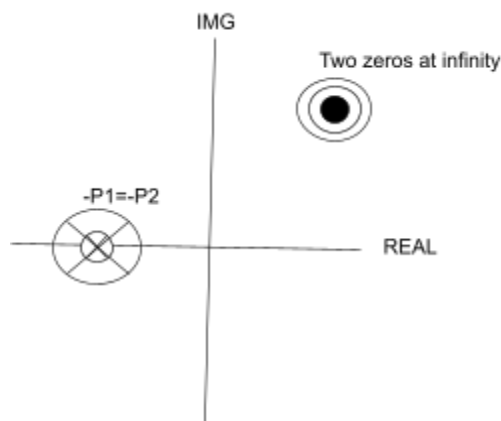
Using the quadratic formula $(\frac{-b \pm \sqrt{b^2 - 4ac}}{2a})$, where $a = 1$, $b = \frac{2}{RC}$, $c = \frac{1}{RC^2}$, we get

$$\frac{-\frac{2}{RC} \pm \sqrt{\frac{2}{RC}^2 - 4(1)(\frac{1}{RC^2})}}{2(1)} = -\frac{1}{RC} \text{ and } -\frac{1}{RC}$$

$$\text{Pole 1: } S1 = -\frac{1}{RC}$$

$$\text{Pole 2: } S2 = -\frac{1}{RC}$$

The poles are negative, identical and real, indicating that we have a stable critically damped system

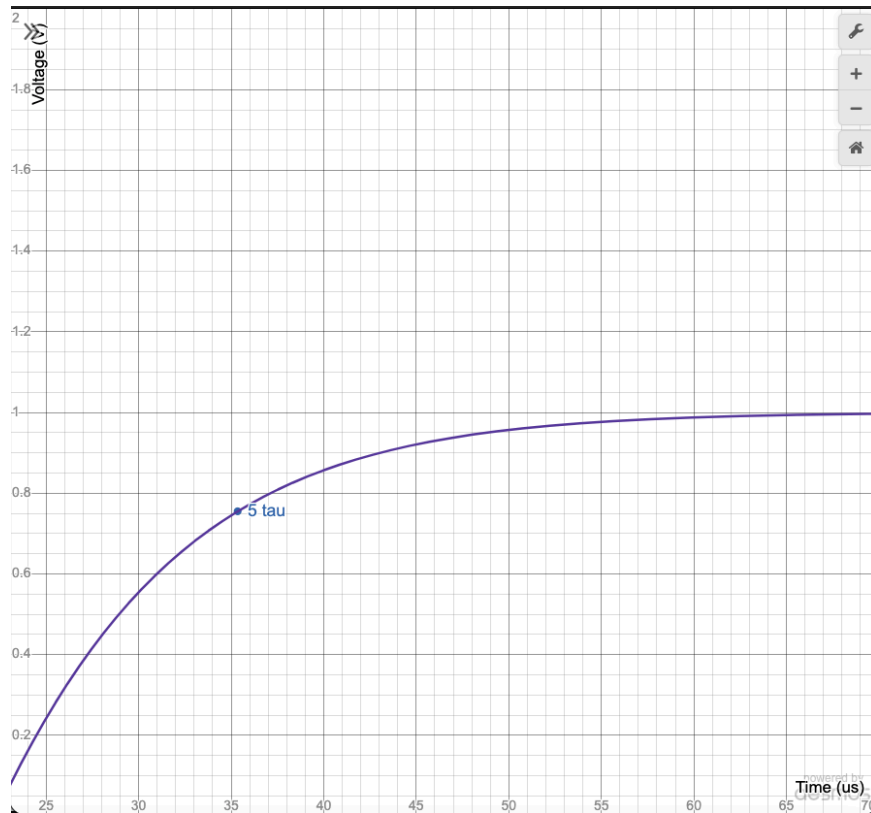


IMG 29.

The step response for a critically damped system is

$$V_o(t) = (1 - [e^{-pt} + te^{-pt}])u(t), \text{ so our step response is}$$

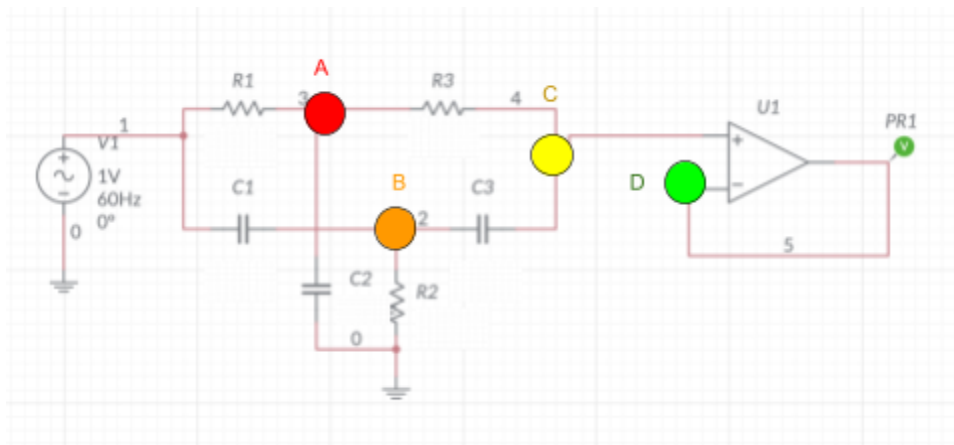
$$V_o(t) = (1 - [e^{-0.1414t} + te^{-0.1414t}])u(t) \text{ and } 5\tau \text{ is } 35.36\mu\text{s at } 0.755\text{V. The process for calculating this can be seen in the previous two circuits.}$$



IMG 30. Steady state is reached after 5τ and V_{max} of 1v is approached, graphed using [Desmos.com](https://desmos.com).

Circuit 4: Twin T Band Stop Filter

Selection of components:



IMG 31.

Node A:

$$\frac{V_a - V_{in}}{R1} + \frac{V_a - V_{ground}}{\frac{1}{C2S}} + \frac{V_a - V_{sc}}{R3} = 0$$

$$\frac{V_a - V_{in}}{R1} + \frac{V_a - 0}{\frac{1}{C2S}} + \frac{V_a - V_c}{R3} = \frac{V_a - V_{in}}{R} + \frac{2CSV_a - 0}{R} + \frac{V_a - V_c}{R} = V_a - V_{in} + 2CSR V_a + V_a - V_c$$

$$(2 + 2CSR)V_a - V_{in} - V_c = 0$$

Node B:

$$\frac{V_B - V_{in}}{\frac{1}{C1S}} + \frac{V_B - V_{ground}}{R3} + \frac{V_B - V_c}{\frac{1}{C3S}} = 0$$

$$V_b C_s - V_{in} C_s + \frac{V_B - V_{ground}}{R/2} + V_b C_s - V_c C_s = R V_b C_s - R V_{in} C_s + 2 V_b + V_b R C_s - V_c R C_s$$

$$(R C_s + 2 + R C_s) V_b - R C_s V_{in} - R C_s V_c =$$

$$(2 R C_s + 2) V_b - (R C_s) V_c - (R C_s) V_{in} = 0$$

Node C:

$$\frac{V_c - V_a}{R2} + \frac{V_c - V_b}{\frac{1}{C3S}} = 0$$

$$\frac{V_c - V_a}{R} + C S V_c - C S V_b = V_c - V_a + R C S V_c - R C S V_b =$$

$$(R C_s + 1) V_c - (R C_s) V_b - V_a = 0$$

Node D:

$$V_d = A V_o$$

Since A=1 due to unity gain

$$V_d = V_o$$

Due to property of an op amp:

$$V_c = V_d, \text{ so } V_c = V_o$$

1) First, find V_a from the equation in terms of v_{out} and v_{in}

$$(RCS + 1)V_c - (RCS)V_b - V_a = 0$$

$$V_a = (RCS + 1)V_o - (RCS)V_b$$

2) Find V_b in terms of v_{out} and v_{in}

$$(2RCS + 2)V_b - (RCS)V_c - (RCS)V_{in} = 0$$

$$V_b = \frac{(RCS)(V_{out} + V_{in})}{(2RCS + 2)}$$

3) Our node C equation can be used to find V_{out} since $V_c = V_o$ now

$$(RCS + 1)V_c - (RCS)V_b - V_a = 0$$

$$(RCS + 1)V_o - (RCS) \left(\frac{(RCS)(V_{out} + V_{in})}{(2RCS + 2)} \right) - \left(\frac{(V_{out} + V_{in})}{(2RCS + 2)} \right) = 0$$

$$x = RCS$$

$$(x + 1)V_o - \left(\frac{x^2(V_{out} + V_{in})}{2(x+1)} \right) - \left(\frac{(V_{out} + V_{in})}{2(x+1)} \right) = 0$$

$$(x + 1)V_o - \left(\frac{x^2 + 1}{2(x+1)} \right)(V_{out} + V_{in}) = 0, (x^2 + 4x + 1)V_o = (x^2 + 1)V_{in}, \frac{V_{out}}{V_{in}} = \frac{x^2 + 1}{x^2 + 4x + 1}$$

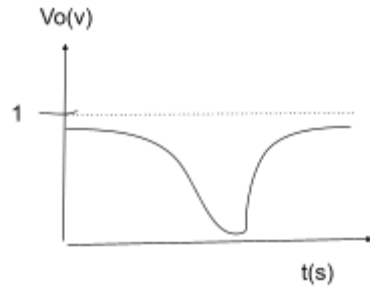
$$\frac{V_{out}}{V_{in}} = \frac{(RCS)^2 + 1}{(RCS)^2 + 4RCS + 1}$$

$$H(s) = \frac{s^2 + \left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{4}{RC}\right)s + \left(\frac{1}{RC}\right)^2}$$

We can see that there is a polynomial expression in the numerator, indicating that this is probably a band stop. To check that this is indeed the transfer function of a band stop filter,

$$\text{As } S \rightarrow 0, H(s) = \frac{s^2 + \left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{4}{RC}\right)s + \left(\frac{1}{RC}\right)^2} \approx \frac{0^2 + \left(\frac{1}{RC}\right)^2}{0^2 + \left(\frac{4}{RC}\right)0 + \left(\frac{1}{RC}\right)^2} \approx 1$$

$$\text{As } S \rightarrow \infty, H(s) = \frac{s^2 + \left(\frac{1}{RC}\right)^2}{s^2 + \left(\frac{4}{RC}\right)s + \left(\frac{1}{RC}\right)^2} \approx \frac{\infty^2 + \left(\frac{1}{RC}\right)^2}{\infty^2 + \left(\frac{4}{RC}\right)\infty + \left(\frac{1}{RC}\right)^2} \approx 1$$



Based on this analysis of the limits of s , the circuit passes low frequency and passes high frequency, thus **behaving like a band stop filter**.

Through comparison of the standard second order equation $\frac{1}{s^2 + (2\xi)\omega_n s + \omega_n^2}$

$$2\xi\omega_n = \frac{4}{RC}$$

$$\omega_n^2 = \left(\frac{1}{RC}\right)^2, \quad \omega_n = \frac{1}{RC}$$

$$2\xi\frac{1}{RC} = 4\frac{1}{RC}, \text{ so } \xi = 2$$

Since $\xi > 1$ the circuit is stable and overdamped

$$\xi = \frac{1}{2Q}, Q=0.25$$

$Q > \frac{1}{2}$	Stable and underdamped
$Q = \frac{1}{2}$	Stable and critically dampened
$Q < \frac{1}{2}$	Stable and overdamped

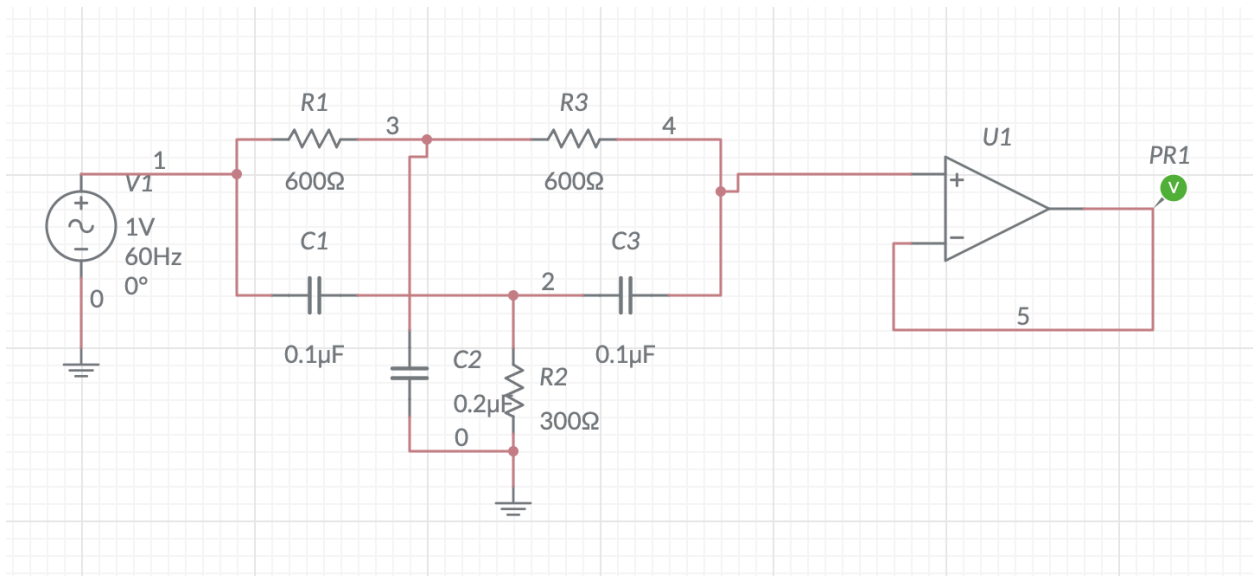
Table 17.

Therefore looking at the quality factor we can see that the low Q indicates a very wide band stop filter, which will be seen in the simulation results.

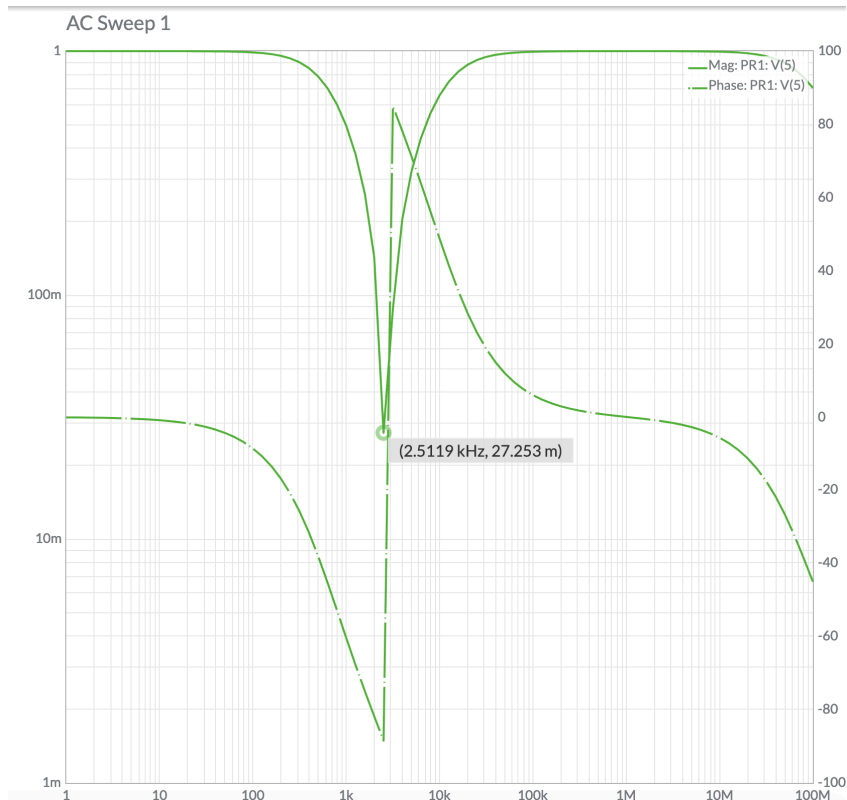
Component	Value	Equivalent
R1	600Ω	R
R2	300Ω	R/2
R3	600Ω	R
C1	$0.1\mu\text{F}$	C
C2	$0.2\mu\text{F}$	2C
C3	$0.1\mu\text{F}$	C

Table 18.

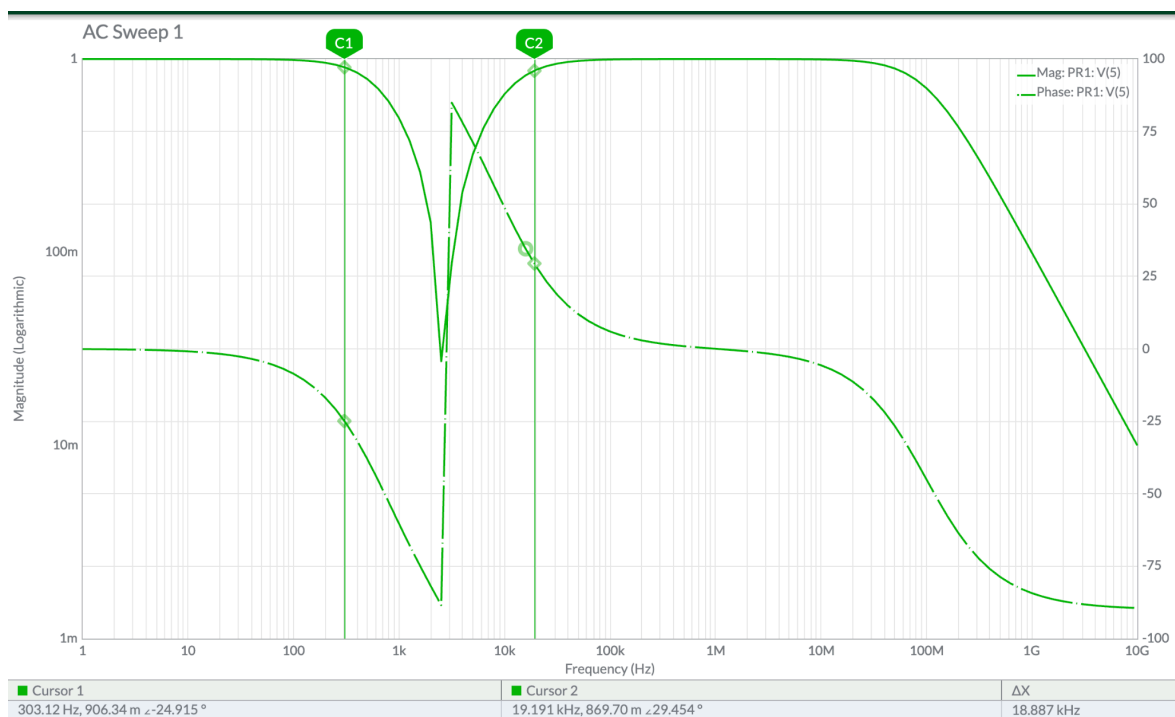
Simulation:



IMG 32. Circuit schematic in Multisim.



IMG 33. Results of Multism schematic. It can be seen that f_c is approximately 2.5kHz



IMG 34. Finding the upper and lower cutoff frequencies in the Multisim.

According to the results from Multisim, $fc = 2.5119kHz$, $fc1 = 303.12Hz$, $fc2 = 19.191k Hz$

check : $fc = \sqrt{fc1 * fc2} = \sqrt{303.12 * 19.191k} = 2.412kHz$, so fc matches and using these frequency values as our bandwidth is reasonable.

Analysis of the poles and zeros:

$$H(s) = \frac{s^2 + (\frac{1}{RC})^2}{s^2 + (\frac{4}{RC})s + (\frac{1}{RC})^2}$$

$$N(Z) = s^2 + (\frac{1}{RC})^2 \text{ and } N(P) = s^2 + (\frac{4}{RC})s + (\frac{1}{RC})^2$$

If $N(z) = 0$, $s^2 + (\frac{1}{RC})^2 = 0$, so $s^2 = -\frac{1}{RC^2}$, $s = \pm j\frac{1}{RC}$. Therefore we have two imaginary zeros at $\pm j\frac{1}{RC}$. On the graph below in image 35 they are shown in the form +/- jb

If $N(P) = 0$, $0 = s^2 + (\frac{4}{RC})s + (\frac{1}{RC})^2$, so we would have to use the quadratic equation

Using the quadratic formula ($\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$), where $A = 1$ $B = \frac{4}{RC}$ and $C = (\frac{1}{RC})^2$

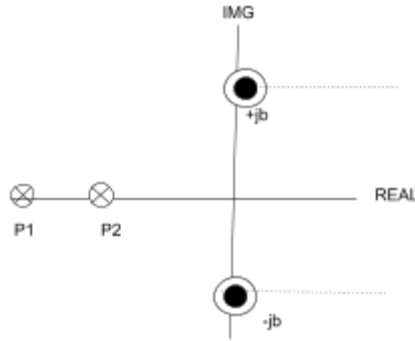
$$\frac{-\left(\frac{4}{RC}\right) \pm \sqrt{\left(\frac{4}{RC}\right)^2 - 4(1)\left(\frac{1}{RC}\right)^2}}{2(1)} = \frac{-\left(\frac{4}{RC}\right) \pm \sqrt{\frac{16}{R^2C^2} - \frac{4}{R^2C^2}}}{2} = \frac{-\left(\frac{4}{RC}\right) \pm \sqrt{\frac{12}{R^2C^2}}}{2} = \frac{-\left(\frac{4}{RC}\right) \pm \frac{\sqrt{12}}{RC}}{2} = \frac{-4 \pm 2\sqrt{3}}{2RC}$$

$$s = \frac{-2 \pm \sqrt{3}}{RC}$$

$$s_1 = \frac{-2 + \sqrt{3}}{RC}$$

$$s_2 = \frac{-2 - \sqrt{3}}{RC}$$

We have two negative, different, real poles which confirms we have an overdamped system



IMG 35.

$$S1 = \frac{-2 + \sqrt{3}}{RC} = \frac{-2 + \sqrt{3}}{600 * 0.1 \mu f} = -4465$$

$$S2 = \frac{-2 - \sqrt{3}}{RC} = \frac{-2 - \sqrt{3}}{600 * 0.1 \mu f} = -62200$$

The step response for an overdamped system is $V_o(t) = [1 - (Ae^{-p1t} + Be^{-p2t})]u(t)$

Our particular step response is $V_o(t) = [1 - (Ae^{-4465t} + Be^{-62200t})]u(t)$

Using partial fractions: $\frac{n(s)}{(s+A)(s+B)} = \frac{A}{s+a} + \frac{B}{s+b}$

$$\frac{s}{(s+4465)(s+62200)} = \frac{A}{(s+4465)} + \frac{B}{(s+62200)}$$

$$n(s) = A(s + 62200) + B(s + 4465)$$

$$\text{Expand: } n(s) = As + 62200A + Bs + 4465B$$

$$\text{Regroup: } n(s) = (A + B)s + (62200A + 4465B)$$

$$62200A + 4465B = 0$$

$$(A + B)s = (1)s, A + B = 1, B = 1 - A$$

$$62200A + 4465(1 - A) = 0, \text{ so } A = -0.0773$$

$$B = 1 - A = 1 - (-0.0773) = 1.0773$$

However, A should be a positive term, and B should be negative. Therefore $A = 1.0773$ and $B = -0.0773$

Therefore, the step response is $V_o(t) = [1 - (1.0773e^{-4465t} - 0.0773e^{-62200t})]u(t)$

Conclusion

This report reviewed the design process for three important topologies commonly used- Sallen Key, Multiple feedback, and Butterworth. In particular the Sallen Key band pass filter, multiple feedback band passfilter, and Butterworth unity gain low pass filter was covered, as well as Twin T band stop filter. The review section covered useful design shortcuts, transfer function derivations, important aspects to be mindful about during component selection, and the purpose of each filter type.

Circuit one was a Sallen Key Band Pass Filter Type 1 with R_f and R_{in} components . Since this circuit was a band pass filter, $F_{c1} \neq F_{c2}$ and transfer function as well as component selection was changed. The signal first passes through the filter consisting of C_2 and R_2 (stage 1) before reaching the op amp, hence the component design and placement of high and low pass filter. The % differences between the calculated and measured values were 7.60% for f_{c1} , 7.87% for f_{c2} , 11.32% for f_c and 11.43% for the bandwidth. This shows that the calculations and theoretical design were reasonably close to the simulated response. The results as well as the pole and zero analysis also confirmed that the circuit was stable and underdamped.

Circuit two was a multiple feedback band pass filter. For this circuit, it was also important to recognize which stage reached the op amp first since this impacts design process and component selection. Nodal analysis used to derive the transfer function for component selection, and then the components were selected based on the desired dampening ratio and frequency response. The simulation gave $f_c = 19.95kHz$, which was close to our theoretical center frequency of f_c of 22.5kHz. Poles, zeros, and underdamped analysis was done, showing that the theoretical design calculations very closely match the results of the simulation.

Circuit three was a Butterworth unity gain low pass filter. Component selection was completed first, and then the transfer function found. Since the filter is a low pass second order filter, it is reasonable to design it with $F_{c1} = F_{c2}$, making the design process and transfer function derivation easier. The Multisim output matched the expected behavior of a second order low pass filter, and the step response was graphed showing that the circuit approached steady state after approximately 5τ . The simulated output matched the intended designed behavior.

Finally, a Twin T band stop filter was designed. Less detail went into this circuit as it is not one of the three critical circuits covered in the review section. Selection of components was done first and then the transfer function was found using nodal analysis. It was shown that the circuit is stable and overdamped. According to the simulation, $fc = 2.5119kHz$, $fc1 = 303.12Hz$, and $fc2 = 19.191kHz$. To check, the geometric mean can be used to find the critical frequency: $fc = \sqrt{fc1 * fc2} = \sqrt{303.12 * 19.191k} = 2.412kHz$, so fc matches and using these frequency values as our bandwidth is reasonable. This circuit had two different negative real poles, confirming that the circuit was overdamped and stable, and had two imaginary zeros. Once again, simulated matched the theoretical calculated results closely.

Overall, the theoretical calculations matched the simulated results reasonably well, confirming that each circuit behaved as intended and that the methodologies used are effective in designing filters and predicting filter behavior.

References:

- **Circuit Analysis in the Frequency Domain & Analog Filter Design**

M. Mousavi, *Circuit Analysis in Frequency Domain & Analog Filter Design*, The Electromechanical System Engineering Technology Department, California State Polytechnic University, Pomona.

- Desmos Studio PBC, “Desmos Graphing Calculator.”

<https://www.desmos.com/calculator>

Accessed: May 1, 2026.